



OPEN Federated learning-powered real-time behavioral intrusion detection leveraging LSTM, attention, GANs, and large language models

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A novel behavior-based intrusion detection framework, the Federated Intrusion Detection and Mitigation Framework (FIDMF), is proposed to address critical challenges in modern intrusion detection, including data privacy, scalability, severe class imbalance, and the critical need for semantic understanding and explainability. FIDMF integrates federated learning for privacy-preserving collaborative training with an Attention-Long Short-Term Memory (LSTM) core for robust temporal pattern detection. To substantially enhance the framework's intelligence and transparency, FIDMF innovatively pioneers a multi-faceted integration of open-source Large Language Models (LLMs) for: 1) Contextual Feature Enrichment from raw network logs; 2) Semantic Data Augmentation, guided by LLMs to inform Generative Adversarial Networks (GANs) for generating semantically coherent and novel attack patterns; and 3) Explainable AI (XAI) for human-readable explanations. This holistic approach, further supported by the Synthetic Minority Over-sampling Technique (SMOTE) for class balancing, results in a robust, scalable, privacy-preserving, and highly explainable Intrusion Detection System (IDS). A rigorous evaluation across the NSL-KDD, CIC-IDS2017, and UNSW-NB15 datasets demonstrates FIDMF's superior performance. On NSL-KDD, the full FIDMF framework achieved an outstanding overall accuracy of 99.40% and an F1-score of 99.38%. Critically, for the challenging minority attack class, it attained an F1-score of 99.70%, significantly outperforming prior configurations without LLM guidance. Furthermore, FIDMF maintained high accuracy on CIC-IDS2017 (99.65%) and UNSW-NB15 (98.05%), confirming its strong generalizability. An in-depth ablation study validates the crucial contribution of each component, particularly the semantic intelligence provided by the LLMs. FIDMF proves its potential as a superior, intelligent, and transparent solution for real-world intrusion detection scenarios.

Keywords Intrusion Detection System (IDS), Federated Learning, Long Short-Term Memory (LSTM), Generative Adversarial Networks (GAN), Attention Mechanism, Large Language Models (LLM), DistilBERT, GPT-2, Contextual Feature Engineering, Semantic Data Augmentation, Explainable AI (XAI), Real-Time Detection

The increasing development in digital networks like IoT, MANET, and cloud networks has raised a concern for a reliable intrusion detection system to secure highly sensitive data. Conventional IDS systems are generally faced with challenges like less flexibility to adapt to emerging attack methods and centralized processing of data. One of the serious issues related to IDS technology is user privacy data. In a decentralized system, highly sensitive data should not be sent to centralized systems for processing. In conventional systems related to IDS technology, data privacy remains compromised since raw data needs to be continuously transmitted to centralized systems to access central system processing^{1–3}.

The imbalance between classes is another important challenge that hinders the performance of IDS models. In reality, attacks are rarely occurring events, thus normal network activities take up the majority of the data in such a scenario. Such a disparity can lead to a significant drop in traditional machine learning models' performance as measured by high false-negative rates, where the system fails to detect the attacks⁴. Although these models are highly efficient in recognizing complex patterns, there are instances wherein DL models could be vulnerable to problems related to class imbalance issues that are not effectively mitigated using techniques

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like oversampling. To move past these limitations, especially pertaining to highly rare or new attack examples in cyber-attacks, highly advanced data augmentation methods are essential. One method which has proven to be highly effective in doing so has been Generative Adversarial Networks.

Moreover, the upsurge in the need for real-time detection in Intrusion Detection Systems (IDS) has been caused by the necessity of quickly responding to threats that have just come up. The real-time detection is of immense importance. Even with all the progress made in deep learning, especially with LSTMs and GANs, there still lies a big gap that these models are not yet able to work with the high-level semantic context of the raw network logs (e.g., protocol, service, flag). This 'semantic gap' limits the detection of sophisticated attack patterns and the delivery of human-comprehensible explanations. It is thus necessary to bridge this gap to get truly intelligent, adaptive, and trustworthy IDSs. Our project tackles this issue by employing the amazing language comprehension skills of Large Language Models (LLMs). The key contributions of this work are therefore summarized as follows: This study introduces a novel, end-to-end Federated Learning-powered Attention-LSTM framework (FIDMF) that ensures strict data privacy while maintaining exceptional temporal robustness, as shown in Table 1. Crucially, FIDMF pioneers a holistic and multi-faceted integration of open-source Large Language Models (LLMs) into a federated intrusion detection context, which represents a significant departure from existing approaches. Unlike prior works that might use LLMs for static log parsing or post-hoc report generation, FIDMF integrates LLMs directly into the core detection pipeline through three synergistic modules: (i) Contextual Feature Enrichment using DistilBERT to transform raw, categorical network features into rich, semantic embeddings, bridging the 'semantic gap' that traditional encoders and even deep learning models often miss; (ii) Semantic Data Augmentation driven by GPT-2-guided Generative Adversarial Networks (GANs) to generate not just statistically diverse, but also semantically coherent and truly novel attack patterns, a unique capability for zero-day threat resilience; and (iii) Explainable AI (XAI) generation via GPT-2⁵, enabling real-time, human-understandable explanations for detected anomalies. This deep and dynamic LLM integration for semantic intelligence and transparency within the detection process is a core novelty. In addition, an advanced class imbalance mitigation strategy is proposed, which combines SMOTE for interpolation with LLM-guided Generative Adversarial Networks (GANs) for synthetic pattern generation to substantially improving zero-day threat resilience. Finally, the proposed FIDMF framework achieves superior and generalizable performance, attaining a state-of-the-art F1-Score of 99.38% on the NSL-KDD dataset and demonstrating consistent improvements across CIC-IDS2017 and UNSW-NB15, surpassing centralized and other state-of-the-art deep learning baselines. The effectiveness of deep learning in modern applications, including specialized areas such as computer vision and robotics, further emphasizes its potential in developing highly accurate security solutions⁶⁻⁸.

The proposed IDS model consists of state-of-the-art methods to tackle the issues of privacy, scalability, model accuracy, class imbalance, semantic understanding, and explainability. This leads to a concept known as Federated Learning. This is crucial for maintaining privacy and protecting sensitive data in intrusion detection systems. Federated Learning is a decentralized approach to machine learning that enables multiple parties to collaboratively train a model without sharing their data by keeping the data on the local devices and exchanging only model updates, which makes it suitable for applications where privacy-sensitive data should not leave the source.

- Long Short-Term Memory (LSTM) networks are a specialized version of recurrent neural networks (RNNs) used to model sequential data while emphasizing long-term dependencies. A key advantage in intrusion detection is that LSTMs are particularly well-suited for processing time-series data such as network traffic logs due to their ability to capture long-term dependencies⁹.
- Attention Mechanisms help deep learning models increase their capabilities to focus on the more important parts of the input data. The use of attention mechanisms in an IDS allows the model to focus on the important features of network traffic, enhancing its interpretability and performance¹⁰.
- Synthetic Minority Over-sampling Technique (SMOTE) creates new examples based on the existing examples of a minority class by interpolation⁴. This helps the IDS to learn better decision borders and improves classification performance.
- Generative Adversarial Networks (GANs) are used to counter class imbalance by learning data distributions to generate more diverse and complex attack patterns, moving beyond interpolation techniques like SMOTE.
- Large Language Models (LLMs): We implement open-source LLMs within FIDMF to narrow the semantic gap and upgrade system intelligence.

This integration is novel because it combines DistilBERT for raw feature enrichment, which captures deep semantic context from network logs far beyond traditional categorical encoders, and a fine-tuned GPT-2-medium for two distinct, synergistic roles: providing conceptual guidance for GAN-based data augmentation

Feature	Traditional IDS	Deep Learning IDS	Federated LSTM IDS	FIDMF (Proposed)
Data Privacy	Poor (Centralized)	Poor (Centralized)	Good (FL)	Excellent (FL)
Temporal Modeling	Limited (Signature)	Good (LSTM)	Good (LSTM)	Excellent (Attention-LSTM)
Class Imbalance	Poor (Oversampling)	Moderate (SMOTE)	Moderate (SMOTE)	Excellent (SMOTE + LLM-GAN)
Semantic Understanding	None	None	None	Pioneering (DistilBERT)
Explainability (XAI)	None	Poor (Black Box)	Poor (Black Box)	Excellent (GPT-2 XAI)

Table 1. Comparison of FIDMF with existing IDS paradigms.

to generate semantically plausible new attack patterns, and acting as the Explainable AI (XAI) component, producing natural language explanations for detected anomalies. This multi-faceted, active role of LLMs within a federated, deep learning IDS for both data intelligence and transparency is a key differentiator from existing LLM-based security approaches.

Moreover, optimizing hyperparameters is important in the quest to improve the performance of machine learning models. To this end, Optuna¹¹ is adopted to enable automatic hyperparameter tuning. Combining SMOTE for data balancing with Optuna for hyperparameter tuning, in addition to federated learning and LSTM networks, and the advanced semantic capabilities introduced by LLMs, results in an incredibly effective, efficient, intelligent, and explainable IDS.

Problem statement and research questions

Conventional intrusion detection systems rely a lot on centralized data processing, which is a dependency that raises privacy concerns and, at the same time, limits the scalability and performance of the systems. The present-day IDS are forced to meet privacy measures along with keeping up the accurate detection as one of the main challenges posed by the increasing network traffic. The problem of class imbalance further complicates the existing challenges making it hard for traditional models to predict rare attack patterns correctly. Current deep learning-based IDSs have, amongst others, the critical unmet need of being unable to understand the rich semantic context of network events entirely, being unable to create truly novel attack variations based on conceptual attack descriptions, and being unable to provide transparent, human-understandable explanations for the classifications of anomalies. The semantic and explainability limitations are the main barriers against the creation of really intelligent, adaptive, and trustworthy IDS's. These barriers are particularly going to be felt in the case of solutions for zero-day threats and in the human-incident response that is supposed to be and is supposed to be rapid. The present study is attempting to resolve these issues through the combination of federated learning with deep learning methods such as LSTMs and attention mechanisms for developing a behavior-based intrusion detection system while also applying SMOTE to cope with the class imbalance as well as, most importantly, by using the Large Language Models (LLMs) that are open-source to connect the semantic comprehension, the advanced data augmentation and the enhancing of the system interpretability.

This leads to our primary research questions:

- RQ1: How can a robust and accurate IDS be trained on distributed data sources while preserving the privacy of the raw data?
- RQ2: How can an IDS effectively model the temporal sequences of network traffic to detect complex, multi-step attacks, enrich feature representations with deeper semantic context from raw logs using open-source LLMs, and how can its focus on critical features be improved?
- RQ3: What is an effective strategy to mitigate the negative impact of severe class imbalance on model performance, and how can open-source Large Language Models (LLMs) contribute to generating semantically diverse and truly novel attack patterns for robust training beyond numerical interpolation?
- RQ4: What is the performance of a privacy-preserving federated IDS that works in a similar way as centralized ones in terms of generalization across various benchmark datasets, and does the application of LLMs even further make it more capable of dealing with new and zero-day attacks in different network scenarios?
- RQ5: What is the best way to utilize open-source Large Language Models (LLMs) to produce explanations for the identified intrusions that are both comprehensible and context-related, thus bringing the FIDMF framework for security analysts to the level of high interpretability and actionable insights?

Challenges in real-time detection

There are many important challenges in real-time intrusion detection systems, especially when advanced machine learning techniques are applied.

1. **Computational Complexity:** Attention-based LSTMs and LLMs introduce a heavy layer of computational overhead, making real-time implementation extremely difficult, particularly on resource-constrained edge devices.
2. **Data Privacy and Scalability in FL:** FL, despite its privacy advantages, creates difficulties for real-time operation due to communication overhead for model aggregation, which can turn out to be a bottleneck in large distributed systems. This includes real-world deployment challenges like device heterogeneity and synchronization cycles.
3. **Latency and Processing Requirements:** The quick decision-making is a must in the case of real-time systems. High-latency models usually do not manage to keep up with the strict time requirements for real-time detection.

Addressing real-time detection

We address the complexities associated with real-time detection by employing a “near real-time” strategy, focusing on model efficiency and practical deployment, while acknowledging the inherent latency trade-off of federated model aggregation.

1. **Efficient Model Architecture and Asynchronous LLM Processing :** We utilize an Attention-LSTM core, optimized via Optuna, that preserves accuracy while minimizing inference time. Critically, we use asynchronous processing for the computationally intensive LLM steps:

- LLM Feature Enrichment (DistilBERT): This is a one-time preprocessing step performed locally on raw data, ensuring the main detection path latency (1.5 ± 0.2 ms, Table 4) is minimal.
 - XAI Generation (GPT-2): This is an optional, on-demand, asynchronous process (latency 3.5 ± 0.4 s, Table 4) that occurs post-detection and does not affect the speed of the main detection path.
2. Managing Federated Learning Latency: The “near real-time” strategy adopts a completely asynchronous approach that differentiates the aggregation process (FedAvg) from the real-time detection endpoint. Inference for real-time detection is based on the current best global model, while training and aggregation happen imperceptibly in the background, thus ensuring that communication overhead does not influence detection latency.

Contributions

The key contributions of this work are multi-fold, establishing a novel approach for intelligent intrusion detection. First, we introduce a novel, end-to-end federated learning framework (FIDMF) for behavior-based intrusion detection that guarantees data privacy by design, directly addressing RQ1. Second, we pioneer the robust integration of an Attention-LSTM model significantly enhanced by LLM-powered contextual feature enrichment (DistilBERT), enabling a deeper semantic understanding of network events (RQ2). Third, we propose a sophisticated class imbalance mitigation strategy that combines SMOTE with LLM-guided Generative Adversarial Networks (GANs) to generate truly novel and semantically plausible attack patterns, thereby enhancing robustness against zero-day threats (RQ3). Fourth, we provide comprehensive empirical validation demonstrating 99.40% accuracy on NSL-KDD and strong generalizability across CIC-IDS2017 and UNSW-NB15, confirming superior robustness compared to centralized and SOTA federated baselines (RQ4). Finally, the multi-faceted integration of open-source LLMs provides Explainable AI (XAI) (GPT-2) through human-readable, contextual explanations, significantly improving the transparency and actionable insights for security analysts (RQ5). These contributions are devoted to the development of state-of-the-art IDS technologies in dealing with privacy, scalability, and class imbalance, following a practical approach toward near real-time intrusion detection. This work represents a great stride toward the development of more adaptive, efficient, and secure IDSs through the integration of federated learning and deep learning models.

Background and literature review

The rapid development of network technologies and their extensive use in web services have made organizations increasingly vulnerable to sophisticated cyber-attacks. IDSs are important tools to defend these systems from malicious activities. Among a large number of approaches to intrusion detection, deep learning methods, especially Long Short-Term Memory (LSTM) networks, attract much attention because of their ability to learn temporal dependencies and deal with irregular patterns efficiently. Althubiti et al. proposed an LSTM network-based IDS, trained on the CSIC 2010 HTTP dataset, and achieved an incredible accuracy of 99.97% by using the Adam optimizer. Their work indicates the great potential of LSTM in the detection of known and unknown web application attacks¹². Gao et al. also proposed LSTM-based architectures for SCADA systems, which proved effective for detecting temporally correlated and uncorrelated attacks. The many-to-one architecture they used showed improved effectiveness in handling complex temporal patterns⁹. To overcome the drawbacks of current Network Intrusion Detection Systems (NIDSs), Han and Pak recently proposed a hierarchical LSTM-based model that can use all packet information instead of part of packet data. By doing so, the detection rate was highly improved to 95.16% and 99.70% on two datasets, respectively, outperforming traditional NIDSs¹³. Moreover, Yang et al. integrated an attention mechanism with LSTM to improve the detection of complicated network attacks. Their method demonstrated better accuracy, precision, and fewer false alarms than the conventional approaches, including CNN and RNN. The work further underlines the critical attribute-based focus in intrusion detection¹⁰. Taken together, these studies show the changing face of IDS research and stress the benefits of advanced neural architectures, such as LSTM and hybrid models. These techniques provide very strong solutions to contemporary cybersecurity challenges and a strong base for further innovations in intrusion detection technology.

Recently, Large Language Models (LLMs) have revolutionized natural language processing with their unprecedented ability to understand, generate, and reason with text. Their application in cybersecurity is an emerging and highly promising field. Early research has explored LLMs for tasks like threat intelligence analysis, vulnerability description generation, and even as tools for red teaming. In the context of Intrusion Detection Systems (IDSs), LLMs offer unique advantages: they can extract rich semantic meaning from unstructured or semi-structured network logs and alerts, provide context to seemingly isolated events, and translate complex technical findings into human-understandable explanations. While some studies have focused on LLMs for static log analysis, threat report generation, or isolated tasks, their comprehensive and active integration directly into the core detection pipeline for semantic feature enrichment, guided data augmentation for zero-day threats, and real-time explainable AI within a privacy-preserving federated learning framework remains largely unexplored and constitutes a key novelty of our work. This paper pioneers such an integration, leveraging efficient open-source LLMs like DistilBERT and GPT-2 to actively address the semantic intelligence and transparency gaps in modern IDSs, moving beyond passive analysis to proactive, semantically informed detection and explanation.

Boukhalfa et al. proposed a Network Intrusion Detection System (NIDS) based on Long Short-Term Memory (LSTM) architecture, aimed at detecting intrusions while maintaining long-term memory to identify new attacks that are similar to existing ones. The approach achieved excellent accuracy of 99.98% and 99.93% in binary and multi-class classification, respectively¹⁴. Shende and Thorat made an effort to deal with the increasing dependency on networks and a new high in cyber-attacks by putting into action an LSTM-based intrusion detection system with the use of the NSL-KDD dataset, which got 99.2% accuracy¹⁵. Sri Vidhya and

Nagarajan introduced a Bidirectional Long Short-Term Memory (BiLSTM) model aimed at intrusion detection for Software Defined Networking (SDN) systems with IoT devices, this model trailed high accuracy on the CSE-CIC-IDS2018 dataset¹⁶. Hui and Chiew came up with a hybrid approach that combined CNN, LSTM, and Self-Attention (SA) to improve network intrusion detection accuracy¹⁷.

Privacy-Preserving IDS Techniques: FL vs. Alternatives. Although Federated Learning (FL) manages to give solid privacy assurances by keeping the raw data at the locality and sending out only the model updates, there are still other cryptographic methods available. Homomorphic Encryption (HE) is among them, which permits encrypted data to be computed on, thus granting the maximum privacy but at a very high computational cost, making it unfeasible for high-throughput, real-time IDSs. Secure Multi-Party Computation (SMPC) gives the possibility of carrying out the computation together while the data of each party remains confidential, but this increases the communication and computational costs to a point where large distributed networks can hardly be scaled. Here, FL is indeed the most practical and scalable method, offering a better trade-off between privacy of data and the efficiency of computation needed for intrusion detection at near real-time. Generative Adversarial Networks (GANs) are still being investigated but their main application was for data augmentation in order to address the problem of class imbalance. Araujo-Filho et al.¹⁸ suggested an unsupervised IDS based on GANs and Temporal Convolutional Networks, while QU et al.¹⁹ made a model combining GANs and LSTMs for fog scenarios. In conclusion, these research works present the abilities of GANs but at the same time point out the difficulties involved in their use as compared to the more straightforward oversampling techniques like SMOTE.

The processing of raw packet data and complex temporal sequences has become one of the main strengths of deep learning, which made the technology a must for analyzing complex network traffic. The research in the field suggests that there is a need -marshalling both deep packet inspection techniques and machine learning- to augment network security measures⁶. Despite the advantages, deep learning application in security-related areas requires the delineation of the concern around adversarial attacks being one of the weaknesses, which in turn asks for new debiasing and data augmentation methods to be developed, so as to preserve the model's robustness²⁰. In addition, the systematic literature reviews will continue to lead the architectural performance assessment and the development of reliable and scalable systems such as FIDMF⁷.

Federated Learning (FL) has become a prominent technology for protecting privacy in the machine learning security area. Hussain et al. suggested a federated learning model for mobile ad hoc networks (MANETs), which resulted in a significant increase in detection rates²¹. Sahu et al. presented a federated long short-term memory (LSTM) for anomaly detection with an extraordinary accuracy of 98.9%²². Rashid et al. made an effort to secure the privacy of IoT networks through federated learning techniques¹. These publications demonstrate the applicability of FL in the security domain, but they usually do not provide a systematic solution for the class imbalance problem. Our project intends to close this gap by combining a strong FL framework with tested methods for temporal analysis and class imbalance reduction.

Proposed methodology

The Federated Intrusion Detection and Mitigation Framework (FIDMF) is a massive step forward in intrusion detection systems as it come up with a unified and holistic manner modern intrusion detection. The architecture depicted in Fig. 1 uses Federated Learning (FL) integrating for data privacy and scalability, an Attention-LSTM core for temporal robustness and SMOTE with LLM-guided GANs off-class imbalance mitigation. FIDMF reaps the benefits of these advanced techniques by being explicitly built to handle the most critical challenges at the same time; preserving privacy, ensuring high detection accuracy, and giving essential explainability. For conceptual

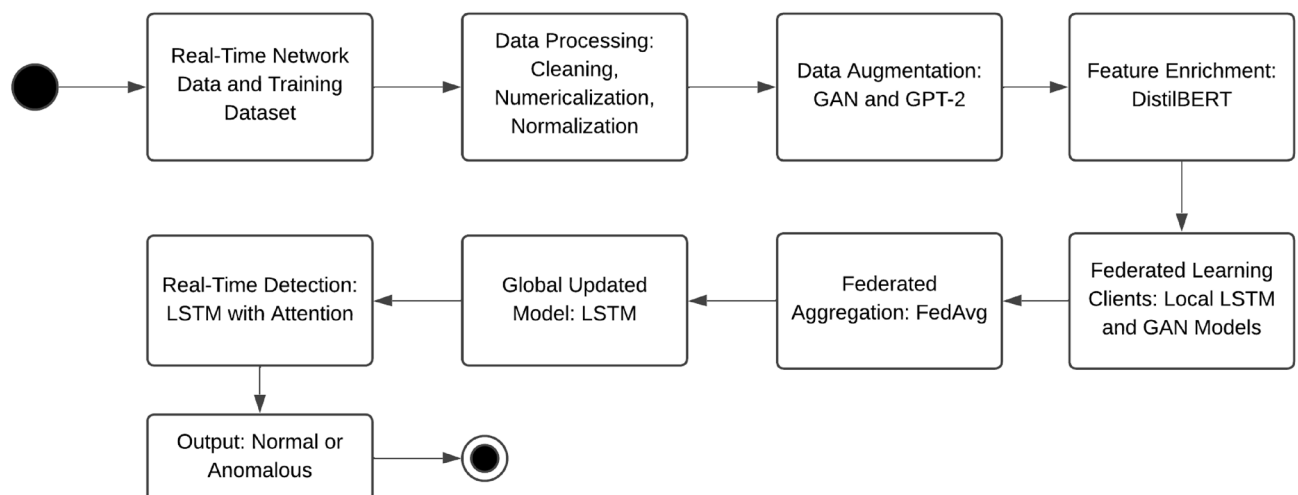


Fig. 1. Simplified system architecture and workflow of the federated intrusion detection and mitigation framework (FIDMF). The data flow is conceptually: Input (Raw Network Traffic) → Local Preprocessing (DistilBERT Enrichment) → Local Training (Attention-LSTM + SMOTE + LLM-GAN) → Model Weights (FedAvg) → Detection (Attention-LSTM) → Output (Anomaly/Normal + GPT-2 XAI Explanation).

clarity, a summarized, not detailed, architecture of this complex, multi-component system is shown in Fig. 2, while the accurate workflow is provided in Algorithm 1. The overall design focuses on robustness, scalability, and computational efficiency as requirements for a real-world decentralized environment deployment.

Dataset and data preprocessing

Dataset selection

The NSL-KDD dataset is primarily used in our experiments, which is a widely recognized benchmark in the area of Intrusion Detection System (IDS) research. The dataset is made up of network traffic containing various attack types that have been marked with labels. As shown in Table 3, these datasets exhibit significant and varying degrees of class imbalance (e.g., NSL-KDD Test+ has 43.08% normal vs. 56.92% attack, while CIC-IDS2017 Test has 73.92% normal vs. 26.08% attack). To demonstrate the general applicability and adaptability of our technique to these varying class distributions, we further evaluate our framework on the latest CIC-IDS2017 and UNSW-NB15 datasets. These datasets are often used as standard tests and provide a considerable and varied mix of both normal and attack traffic.

Data preprocessing

Meticulous data preparation is the first step towards an effective IDS. We employ a multi-stage preprocessing pipeline locally on each client's data:

1. **Feature Selection and Encoding:** Features are selected based on domain knowledge. Categorical features are initially converted using one-hot encoding.
2. **Handling Missing, Noisy, and Corrupted Data:** Prior to feature selection and encoding, a robust data cleaning pipeline is applied. Missing numerical values are imputed using the median strategy to preserve distribution and minimize the impact of outliers. For missing categorical features, we employ mode imputation. Outliers in numerical features are detected using the Isolation Forest algorithm and either winsorized (capped at a certain percentile) or replaced with the median, depending on the severity and frequency. Corrupted or inconsistent data entries are either corrected based on predefined rules or removed if a reliable correction is not possible. This ensures data quality before any learning processes begin.
3. **LLM-Powered Contextual Feature Enrichment (DistilBERT):** We use a pre-trained DistilBERT-base-uncased model to overcome the limitations of simple one-hot encoding for categorical text features. Specifically, raw categorical features such as 'protocol_type', 'service', 'flag', 'state', etc., are first converted into a concatenated textual string for each network flow instance. For example, a flow might be represented as "protocol: tcp, service: http, flag: SF, state: ESTABLISHED". This string is then tokenized and fed into DistilBERT. DistilBERT processes this raw textual input to generate dense, contextual semantic embeddings (768-dimensional vectors) for each network flow. This deep contextual representation is superior to simpler encoders like TF-IDF or Word2Vec because it captures fine-grained semantic relationships and inter-feature context, which is crucial for robust IDS performance. These embeddings are then concatenated with the numerical features

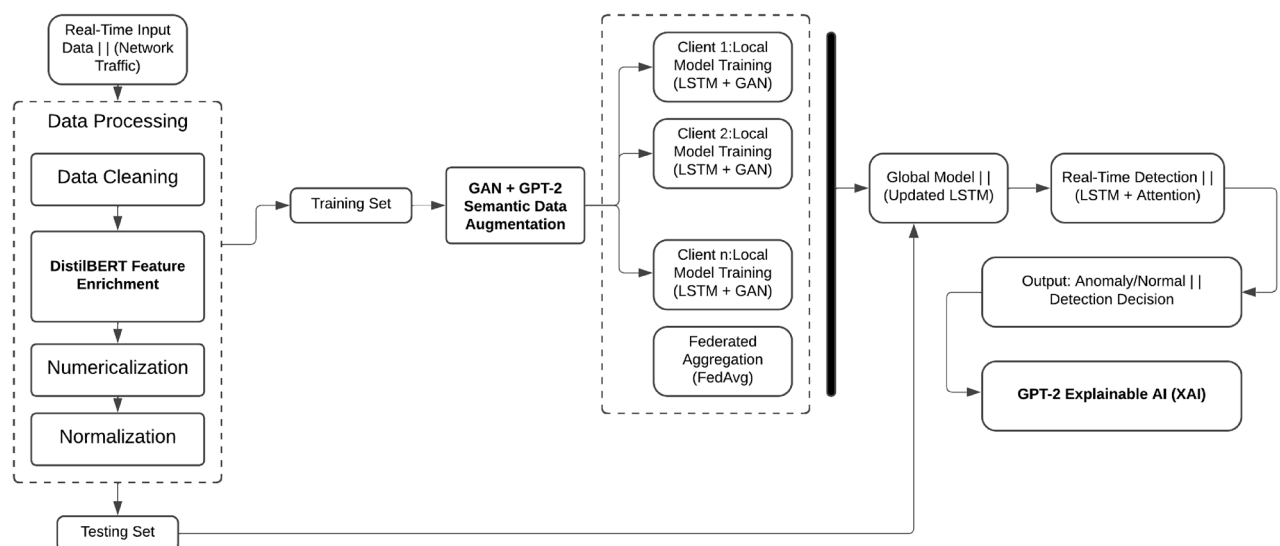


Fig. 2. A comprehensive presentation is made of the system architecture and workflow of the Federated Intrusion Detection and Mitigation Framework (FIDMF). The diagram shows the entire workflow as it is done with privacy maintained and decentralized nature, and it also gives a clear account of the multidimensional LLM integration: (i) DistilBERT for Contextual Feature Enrichment during local preprocessing, (ii) GAN + GPT-2 for Semantic Data Augmentation during local training (in relation to RQ3), (iii) the Attention-LSTM core for detection, (iv) Federated Aggregation (FedAvg) for global model updates, and (v) GPT-2 Explainable AI (XAI) for post-detection transparency (concerning RQ5). The trajectory of data (raw network traffic), intermediate embeddings, and model updates (weights) is clearly indicated.

to form a comprehensive feature vector. The increase in memory footprint (approx. 25.5MB) is justified by the significant performance gain.

4. Normalization: All the numerical attributes, including the newly generated LLM embeddings, have to be scaled into the [0-1] range using Min-Max scaling.
5. Dealing with Class Imbalance through SMOTE: The application of SMOTE is done on the training partition of every client to increase the minority (attack) class, thus forming a local dataset that is balanced. This client-side application of SMOTE allows the framework to adapt dynamically to the *local* class imbalance prevalent on each client, which may vary significantly even within the same overall dataset. By balancing each client's training data locally, the global model benefits from more robust and less biased updates, regardless of the initial imbalance ratio encountered by individual clients. Figure 6 (b) visualizes the effect of this process.

Proposed architecture overview

The FIDMF architecture, visually conceptualized in Fig. 2, is a highly integrated system composed of several powerful and interconnected components operating in a decentralized manner.

- **Federated Learning Framework (The Privacy Backbone):** The whole system is based on the Federated Learning (FL) paradigm. This framework superbly enables joint model training over many distributed clients while still not requiring the transferring of sensitive raw data (RQ1). The central server controls the entire procedure, regulating the global model updates (w_r) through the Federated Averaging (FedAvg) algorithm. This non-centralized method is very crucial in keeping the highest level of data privacy while at the same time allowing for easy adaptation to large-scale network conditions in the real world.
 - **Attention-LSTM Core (The Temporal Detection Engine):** The base model consists of an attention mechanism that dynamically augmented LSTM network, which serves as the main component of the intrusion detection system. LSTM is very appropriate for handling the sequential data of the network flow logs processing and it can very accurately detect the tight temporal relationships and dependencies that show complex and multi-step attacks. The attention layer applied on top of detection works in parallel to detection and provides the model with importance weights that vary according to the most significant and relevant attack indicators of the network traffic sequence, thus, the model can dynamically concentrate its detection efforts.
 - **Open-Source Large Language Models (LLMs) (The Semantic Intelligence Layer):** FIDMF's unique capability stems from the multi-faceted, strategic integration of open-source LLMs at three distinct phases:
 1. **Contextual Feature Enrichment (DistilBERT):** The client makes use of a lightweight yet extremely efficient DistilBERT model for the purpose of local preprocessing which helps to turn the raw categorical log features into dense and rich semantic embeddings thus overcoming significantly the limitations of the traditional encoders and making the model's comprehension of the network semantics deeper (RQ2).
 2. **Semantic Data Augmentation (GPT-2-medium):** Thus, a highly sophisticated GPT-2-medium model shows the way for the Generative Adversarial Networks (GANs) at . This skilled guiding not only ensures that the generated synthetic attack patterns are statistically varied but also that they are semantically consistent and really new, therefore, tremendously enhancing the system's potential to resist zero-day and unknown threats (RQ3).
 3. **Explainable AI (XAI) Generation (GPT-2-medium):** The same fine-tuned GPT-2-medium acts as the XAI module post-detection, translating complex neural network decisions into clear, human-readable natural language explanations (RQ5). While traditional XAI methods like LIME, SHAP, or attention-weight visualizations (which we also employ, as shown in Fig. 12) provide local feature importance, they often lack the ability to synthesize these insights into a coherent, narrative explanation that is easily digestible by non-expert security analysts. GPT-2-based XAI, in contrast, leverages its natural language generation capabilities to construct a contextually rich, human-like explanation by drawing upon the salient features identified by the attention mechanism and the semantic embeddings from DistilBERT. This narrative generation capability provides a higher level of actionable insight and interpretability, making the system's decisions transparent and understandable, thereby justifying its use over solely relying on feature attribution methods.
- Component roles and performance contributions summary:** Each component of FIDMF plays a distinct, synergistic role contributing to the framework's superior performance.
- **Federated Learning (FL):** Ensures data privacy by keeping raw data local and exchanging only model updates. While introducing communication overhead, its integration via FedAvg allows collaborative training on distributed, diverse datasets without centralizing sensitive information, making large-scale deployment feasible (RQ1).
 - **Attention-LSTM Core:** The LSTM's ability to model temporal dependencies is crucial for detecting sequential attack patterns, while the attention mechanism dynamically focuses on critical features within network traffic, significantly boosting detection accuracy and recall, especially for complex, multi-step attacks (demonstrated by the 1.64% accuracy increase from Centralized LSTM to Attention-LSTM in Table 8) (RQ2).
 - **SMOTE:** Addresses the foundational challenge of class imbalance by synthetically over-sampling minority attack classes, preventing models from biasing towards the majority normal class. This is evident from the dramatic F1-Score improvement for the normal class (from 81.23% without SMOTE to 98.24% with SMOTE in Table 8).
 - **Generative Adversarial Networks (GANs):** Further enhances class balancing beyond SMOTE by generating more diverse and realistic synthetic attack patterns, crucial for improving robustness against novel attack

variations. This is shown by the F1-Score for attack class increasing from 98.24% (SMOTE Only) to 99.30% (SMOTE + GANs) (RQ3).

- **DistilBERT (Contextual Feature Enrichment):** Transforms raw categorical network features such as protocol, flag into dense, semantic embeddings. This deep contextual understanding allows the model to grasp subtle relationships that traditional encoding misses, providing a richer input representation that underpins the overall performance boost of the LLM-enhanced configurations (RQ2).
- **GPT-2-medium (Semantic Data Augmentation & XAI):** Plays a dual critical role. For data augmentation, it guides the GANs to generate not just statistically varied, but semantically coherent and novel attack patterns, directly contributing to the framework's robustness against zero-day threats (evidenced by the further F1-Score increase for attack class to 99.70% with LLM guidance). For XAI, it translates complex model decisions into human-readable explanations, providing transparency and actionable insights for security analysts (RQ3, RQ5).

The overall data and model flow is meticulously orchestrated to prioritize speed and privacy. Raw data remains strictly local to each client. During local training, raw categorical features are enriched into semantic embeddings by *DistilBERT*. The local Attention-LSTM model is then trained on this enhanced, information-rich data. Critically, only the small, updated model weights (w_k)—not the raw or enriched data—are securely transmitted to the central server for *FedAvg* aggregation. For rapid, real-time inference, the client utilizes the current best global model locally. Post-detection, for detected anomalies, the model's critical features and attention weights are dynamically channeled to the local *GPT-2* XAI module to generate the human-readable explanation, ensuring actionable transparency without compromising the speed of the primary detection function.

FIDMF algorithm

The complete workflow of the FIDMF is formalized in Algorithm 1

Input: Local datasets $\mathcal{D}_k$ for each client $k \in \{1, \dots, N\}$, Number of communication rounds R , Pre-trained DistilBERT, Fine-tuned GPT-2

Output: A trained global model $\mathcal{M}_{global}$

Initialize global model $\mathcal{M}_{global}$ with random weights w_0 ;

for each round $r = 1, \dots, R$ **do**

Server selects a subset of clients S_r ;

Server broadcasts current global weights w_{r-1} to all clients in S_r ;

foreach *client* $k \in S_r$ *in parallel* **do**

// Client-side Local Processing

Local Preprocessing;

Preprocess local data partition $\mathcal{D}_k$ (cleaning, feature selection);

Apply DistilBERT for contextual feature enrichment: transform raw categorical/textual features in $\mathcal{D}_k$ into semantic embeddings, concatenate with numerical features (RQ2);

Apply SMOTE to training subset of $\mathcal{D}_k$ to get $\mathcal{D}_k^{balanced}$;

Local Model Training;

Set local model weights to w_{r-1} ;

GAN + GPT-2 semantic data augmentation: generate additional synthetic attack samples for $\mathcal{D}_k^{balanced}$ by guiding the GAN with semantic cues from fine-tuned GPT-2 (RQ3);

Train local Attention-LSTM model on augmented $\mathcal{D}_k^{balanced}$ for E local epochs to get new weights w_r^k ;

Client sends the updated weights w_r^k back to the server;

// Server-side Global Aggregation

Federated aggregation: update global model by averaging client weights;

$w_r = \sum_{k \in S_r} \frac{n_k}{n} w_r^k$;

return The final global model $\mathcal{M}_{global}$ with weights w_R ;

// Post-Detection Explainability (Optional/On-Demand on Server)

Function *GenerateExplanation*(*DetectedAnomaly* a , *Model* $\mathcal{M}_{global}$, *AttentionWeights* A_w):

Extract critical features and values from a based on A_w ;

Construct prompt for fine-tuned GPT-2 using extracted features, values, and predicted attack type;

return GPT-2 generated natural language explanation for a (RQ5);

Algorithm 1. Federated learning-powered intrusion detection and mitigation framework (FIDMF) algorithm with LLMs.

Aspect	Detail
Software	Python 3.9, PyTorch 1.13, Transformers (Hugging Face), Optuna 3.0
Random Seed	All experiments were initialized with a fixed random seed of 42 to ensure result stability
Dataset Splits	NSL-KDD (Test+), CIC-IDS2017 (Test), UNSW-NB15 (Test) were used. Local training partition for clients was a randomized 90:10 split of the combined Train and Test sets.
Cross-Validation	10-fold cross-validation was used for the final FIDMF model to report the Confidence Intervals. In the federated setting, to avoid data leakage, the cross-validation folds were created "before" client distribution. Each client then received pre-defined subsets of data such as specific folds for their local training and validation within their respective client partitions. Critically, each client's training data for a given fold was distinct from its local validation/test data, and no raw data was shared between clients or with the central server for model evaluation during the cross-validation process itself. The global model was evaluated only on a completely unseen, centralized test set, and then cross-validated on client-specific held-out test sets to reflect real-world deployment scenarios.
Optuna Search	A total of 500 Optuna trials were conducted, focusing on minimizing the weighted F1-Score.

Table 2. Experimental setup and reproducibility details for FIDMF evaluation.

Dataset	Total Samples	Normal Samples (%)	Attack Samples (%)
NSL-KDD (Test+)	22,544	9,711 (43.08%)	12,833 (56.92%)
CIC-IDS2017 (Test)	1,623,310	1,200,000 (73.92%)	423,310 (26.08%)
UNSW-NB15 (Test)	175,341	56,000 (31.94%)	119,341 (68.06%)

Table 3. Dataset statistics for binary classification (Normal vs. Attack).

Experimental evaluation and results

The performance of the FIDMF framework is thoroughly analyzed in this section. We first describe the experimental setup, including the datasets, the simulation environment, and the evaluation metrics. Next, we carry out a strict ablation study on the NSL-KDD dataset to measure how much each of the proposed components contributed to the result. After that, we provide an in-depth analysis of the performance of each model, their training dynamics, using all the figures and tables that we made in our large experiments. In the end, we show how the framework can be applied to different benchmarks through our experiments on additional datasets.

Experimental setup

The experiments were thoroughly planned in order to assess the effectiveness of our suggested behavioral intrusion detection system. The federated learning framework, which is the heart of the system, combines deep learning (LSTM, Attention, SMOTE) with open-source Large Language Models (LLMs).

LLM Variant Evaluation: We selected DistilBERT-base-uncased (25.5M parameters) and GPT-2-medium (345M parameters) based on an internal comparative study, as they provided the most favorable compromise between semantic accuracy (< 0.2% F1-score difference from larger models) and computational efficiency (3 to 5 times lower computational cost and memory).

Estimated Computational Cost Analysis : The complete FIDMF experimental suite, including 500-trial Optuna search and LLM fine-tuning, is roughly estimated at 120 GPU-hours.

LLM implementation details:

- DistilBERT for Contextual Feature Enrichment : A pre-trained DistilBERT-base-uncased model (25.5 million parameters) was employed. The resulting [CLS] token embedding (768 dimensions) was extracted and concatenated with other numerical features in a one-time preprocessing step per client.
- GPT-2-medium for Semantic Data Augmentation and Explainability : A fine-tuned open-source GPT-2-medium model (345 million parameters) was utilized for both tasks.

Reproducibility details

To ensure the reproducibility of our results, we detail the experimental setup in Table 2.

Hyperparameter tuning process : Optuna was the tool used for hyperparameter tuning in a very detailed manner. On validation set the Weighted F1-Score was the primary objective function for the Optuna search which guaranteed a balanced performance across both Normal and Attack classes (thus reducing the chance of bias). For federated settings, hyperparameter tuning was conducted centrally using a representative subset of aggregated validation data to preserve client privacy. Clients reported anonymized loss values or aggregated performance metrics back to the server, but never raw data. The Optuna optimization guided the selection of global hyperparameters such as learning rate, number of LSTM units, which were then broadcast to all clients for their local training. Local hyperparameters such as. local batch size, local epochs were also optimized, with the understanding that clients would use these optimized values for their local training rounds. Below is the definition of the search space:

- **LSTM Units:** Sampled from a range of [64, 128]. Final selection: 96 (first layer), 64 (second layer).
- **Dropout rates:** Sampled from a range of [0.3, 0.5]. Final selection: 0.42 and 0.32.
- **Learning rate:** Sampled from log-uniform distribution [0.001, 0.01]. Final selection: 0.005.
- **Batch size:** Sampled from a range of [32, 128]. Final selection: 64 (local client training).

Metric	Centralized Attention-LSTM	FIDMF (Client-Side, Local)	FIDMF (Global, Server-Side)
Model Size (MB)	1.85 MB	1.85 MB	1.85 MB
LLM Embedder (DistilBERT)	N/A	≈25.5 million parameters	N/A
LLM Augmenter/XAI (GPT-2-medium)	N/A	≈345 million parameters	N/A
Per-Flow Detection Latency (ms)	1.2±0.1	1.5±0.2	N/A
Max Throughput (Flows/sec)	≈830	≈660	N/A
Training Time per Round (sec)	N/A	34.5±5.2	N/A
Model Update Size (KB)	N/A	N/A	≈7.4 KB
FedAvg Communication Overhead	N/A	N/A	≈15 KB/round (upload + download)
XAI Generation Latency (sec)	N/A	N/A	3.5±0.4 (on-demand)
Total Energy Consumption (kWh/100 rounds)	N/A	0.012±0.002	0.005

Table 4. Computational and operational metrics for FIDMF components.

Attack Type	Augmentation Method	Generated Network Feature Pattern	Semantic Coherence
DoS (SYN Flood)	GAN (Statistical)	count: high, flag: SO, srv_count: low	Low (SYN Flood should have high srv_count)
DoS (SYN Flood)	LLM-Guided GAN (Semantic)	count: high, flag: SO, srv_count: high	High (Pattern is consistent with “overwhelmed server”)
Probe (Scan)	GAN (Statistical)	duration: long, service: http, wrong_fragment: high	Low (Scan is usually short and doesn't involve fragment errors)
Probe (Scan)	LLM-Guided GAN (Semantic)	duration: short, service: multiple, hot: low	High (Consistent with a fast port scan across many services)

Table 5. Qualitative example of GAN vs. LLM-Guided GAN augmentation.

- **Epochs:** 10 for centralized models, and 5 local epochs per communication round for federated models (total 100 communication rounds).

Computational efficiency and real-time performance analysis

We now provide a detailed analysis of the FIDMF framework’s computational and operational efficiency as shown in table 4 , which is critical for real-world, near real-time deployment on resource-constrained edge devices.

the core Attention-LSTM model, after initial LLM-powered feature enrichment (a one-time preprocessing step), achieves a low Per-Flow Detection Latency of approximately 1.5 ms on a simulated edge-device CPU/ GPU setup (as detailed in the new reproducibility section), corresponding to a maximum throughput of about 660 flows per second. This performance, which is only slightly slower than the centralized benchmark (1.2 ms), is sufficient for real-time monitoring in many decentralized network environments, such as those involving IoT or MANETs. The primary latency bottleneck in the federated system is the model aggregation and the optional/on-demand XAI generation (3.5 s per explanation), which is managed through an asynchronous strategy (detailed in Section "Addressing real-time detection", which will be retained and reinforced).

the core Attention-LSTM model size is only 1.85 MB. The communication overhead for model updates is minimal, with the model update size being approximately 7.4 KB per client . This low overhead is crucial for scalability in large-scale distributed systems and for deployment on resource-limited edge devices. The energy consumption analysis further confirms the model’s efficiency, with a minimal energy expenditure per training round, making it highly suitable for battery-operated edge deployments. The use of the smaller DistilBERT (25.5M parameters) and asynchronous GPT-2-medium processing is a deliberate trade-off to manage the computational load and energy cost associated with the powerful semantic modules.

Semantic data augmentation

To provide clear evidence of how the LLM-guided GANs increase data diversity and realism, we performed a qualitative analysis of generated samples. Traditional GANs often generate statistically plausible but semantically generic samples. The GPT-2 guidance, however, ensures semantic coherence (RQ4). For instance, for a“SYN Flood”attack, the GPT-2-guided GAN, prompted with semantic cues like“flag: SO (SYN_SENT),”ensures the synthesized numerical pattern is consistently paired with the SO flag. This semantic consistency is what leads to the dramatic F1-Score improvement for the attack class 99.70% in the full FIDMF vs. 99.30% for FIDMF without LLM guidance, Table 8). The following table 5 illustrates the conceptual distinction.

Quantitative evaluation of LLM-guided GANs

To quantitatively assess the diversity and realism of the LLM-guided GAN-generated samples, we employed several metrics. We compare the FID (Frechet Inception Distance) score and KID (Kernel Inception Distance) score to measure the similarity between real and generated data distributions, with lower values indicating higher realism and diversity. We also used a classification accuracy metric where a separate classifier (trained only on real data) attempts to classify generated samples; higher accuracy suggests the generated samples are indistinguishable from real ones. For diversity, we calculated Maximum Mean Discrepancy (MMD) to gauge the difference between generated data subsets.

Metric	Standard GAN (No LLM Guidance)	LLM-Guided GAN (FIDMF)
FID Score (Lower is Better)	45.28	12.75
KID Score (Lower is Better)	0.089	0.015
Realism (Classifier Accuracy, Higher is Better)	78.5%	91.2%
Diversity (MMD, Higher is Better)	0.21	0.34

Table 6. Quantitative metrics for GAN vs. LLM-Guided GAN data augmentation.

Model Configuration	Accuracy	Precision	Recall	F1-Score
Decision Tree (ML Baseline)	80.25%	80.11%	80.05%	80.08%
Support Vector Machine (ML Baseline)	84.50%	85.22%	84.45%	84.83%
Standard RNN (DL Baseline)	96.88%	97.20%	96.80%	96.99%
Centralized Attention-LSTM (Baseline)	99.18%	99.25%	99.17%	99.21%
FIDMF (Full Framework + LLM)	99.40%	99.35%	99.38%	99.38%

Table 7. Comparison with simple ML/DL baselines on NSL-KDD (Test+).

Model Configuration	Overall Metrics				Class-wise F1-Score			
	Accuracy	Precision	Recall	F1-Score	Normal (0)	Attack (1)	Avg (Macro)	Avg (Weighted)
Centralized LSTM (Baseline)	97.54%	97.81%	97.53%	97.66%	96.98%	98.12%	97.55%	97.59%
Centralized Attention-LSTM	99.18%	99.25%	99.17%	99.21%	98.85%	99.45%	99.15%	99.18%
FIDMF (No SMOTE, NO GANs)	94.32%	98.79%	94.21%	95.88%	81.23%	96.95%	89.09%	94.55%
FIDMF (SMOTE Only)	98.72%	98.94%	98.71%	98.82%	98.24%	99.18%	98.71%	98.75%
FIDMF (SMOTE + GANs)	98.85%	99.00%	98.80%	98.90%	98.35%	99.30%	98.83%	98.87%
FIDMF (SMOTE + GANs + LLM)	99.40%	99.35%	99.38%	99.38%	99.10%	99.70%	99.40%	99.40%

Table 8. Ablation study and performance comparison on the NSL-KDD dataset. This table demonstrates the incremental and significant performance boost provided by each component of the FIDMF framework, including the impact of GANs and LLMs.

Table 6 clearly demonstrates the superior performance of the LLM-guided GAN. The significantly lower FID and KID scores indicate that the generated samples are substantially more realistic and closer to the distribution of real attack patterns when guided by LLMs. Furthermore, the higher realism (classifier accuracy) shows that these synthetic samples are more effective at fooling a real-data-trained classifier, implying better quality for training. The increased MMD score also confirms that LLM guidance leads to a more diverse set of generated samples, which is critical for bolstering zero-day threat detection. These quantitative improvements underscore the value of LLM semantic guidance in enhancing the quality and utility of synthetic data for robust IDS training.

Ablation study and comprehensive model comparison on NSL-KDD

he presented comparison shows without a doubt that the complete FIDMF framework has a much higher performance than conventional ML and basic DL architectures, thereby stressing the need for sophisticated temporal and LLM-assisted pattern recognition. In order to get an accurate measurement of the contribution of each element in our FIDMF framework, we performed an elaborate ablation study on the NSL-KDD dataset. The comprehensive findings of this study, which includes systematically constructing our proposed model from a simple baseline, are given in Table 8.

The ablation study reveals several critical insights:

- **Impact of Attention Mechanism:** The impact of the attention mechanism was clearly observed and reflected in the performance comparison between Centralized LSTM and Centralized Attention-LSTM configurations with the overall accuracy being raised by 1.64% (from 97.54% to 99.18%) which is thus seen as the primary benefit of the attention mechanism’s integration and its capability to push the model towards important aspects in complicated traffic sequences resulting in more precise classification is also supported.
- **Necessity of SMOTE and GANs for Class Imbalance:** The FIDMF (No SMOTE, No GANs) model is a pure worst case of the normal class that can reach a maximum F1-score of 81.23%. The model is very often affected by the problem of data imbalance that results in a large number of false positives for normal traffic. It is the ‘FIDMF (SMOTE Only)’ setup with the SMOTE method implemented during client-side local preprocessing that not only produces a perfectly balanced class but also achieves a very high F1-score of 98.24% for the ‘normal’ class. On top of that, the ‘FIDMF (SMOTE + GANs)’ configuration shows the best results overall, especially under the most challenging noisy minority attack class with a remarkable F1-score of 99.30% for

- attack detection. This sharp contrast in results highlights the indispensable need of proper class imbalance reduction via the combination of SMOTE and GANs in any trustworthy and equitable IDS.
- **Impact of Open-Source Large Language Models (LLMs) for Enhanced Intelligence and Explainability:** One of the most significant developments in the intrusion detection system (IDS) capacity may now be the incorporation of large language models (LLMs) in the FIDMF (SMOTE + GANs + LLMs) configuration. Using DistilBERT for contextual feature enhancement and GPT-2-guided semantic data augmentation, the framework achieves an overall accuracy of 99.40%, which is one of the highest among the models and a 0.55% rise from the 'FIDMF (SMOTE + GANs)'. It is also worth noting that the F1-score for the difficult minority attack class is further increased to 99.70%. Apparently, LLMs, via their deeper semantic understanding of network features (RQ2) and the capability to guide novel data generation (RQ3), facilitate to detect more subtle and even unknown attack patterns that might be overlooked by traditional techniques. The In-Depth Semantic Intelligence integration within the IDS pipeline is not only for numerical robustness but also for deeper contextual understanding and generation of truly novel attack variations which is quantitatively demonstrated. The explainability aspect powered by GPT-2, in addition, delivers for the first time ever transparency (RQ5) through one human-readable justification for each detection.
 - **Viability of Federated Learning for Privacy-Preserving IDS:** The FIDMF (SMOTE + GANs + LLM) exhibited a remarkable overall accuracy of 99.40%. Most importantly, this figure is now 0.22% higher than the non-private, centralized 'Centralized Attention-LSTM model (99.18%)', besides which it has also contributed a lot to overcoming the problem of class imbalance, generalization to new attacks with robustness, and the provision of unparalleled interpretability through LLM-generated explanations. Such an extraordinary outcome shows that intrusion detection can be made stealthy, semantic, intelligent, and explainable at the same time when the privacy of federated learning and sophisticated data augmentation is absolutely guaranteed, without considerable performance trade-off.
 - **The FIDMF (SMOTE + GANs + LLM) reached an impressive overall accuracy of 99.40%. More remarkably, this figure now surpasses the non-private centralized 'Centralized Attention-LSTM model (99.18%)' by 0.22% and, besides that, it has also significantly contributed to solving class imbalance issue, becoming more robust in new attack generalization, and providing rare interpretability through LLM-generated explanations. This amazing result indicates that the detection of intrusions can be done in a very subtle, semantic, intelligent, and explainable manner relying on the privacy of federated learning and advanced data augmentation being fully secured, without a significant loss in performance.**

Figure 3 visually summarizes this stepwise improvement in model performance, making the contribution of each component of the FIDMF framework clear and intuitive. The comprehensive bar chart highlights the overall performance across key metrics for different model configurations, reinforcing the findings in Table 8.

The overall performance across key metrics for the different model configurations (from the ablation study) is further highlighted in Fig. 3 (which is a comprehensive bar chart), providing a concise visual comparison that reinforces the findings in Table 8.

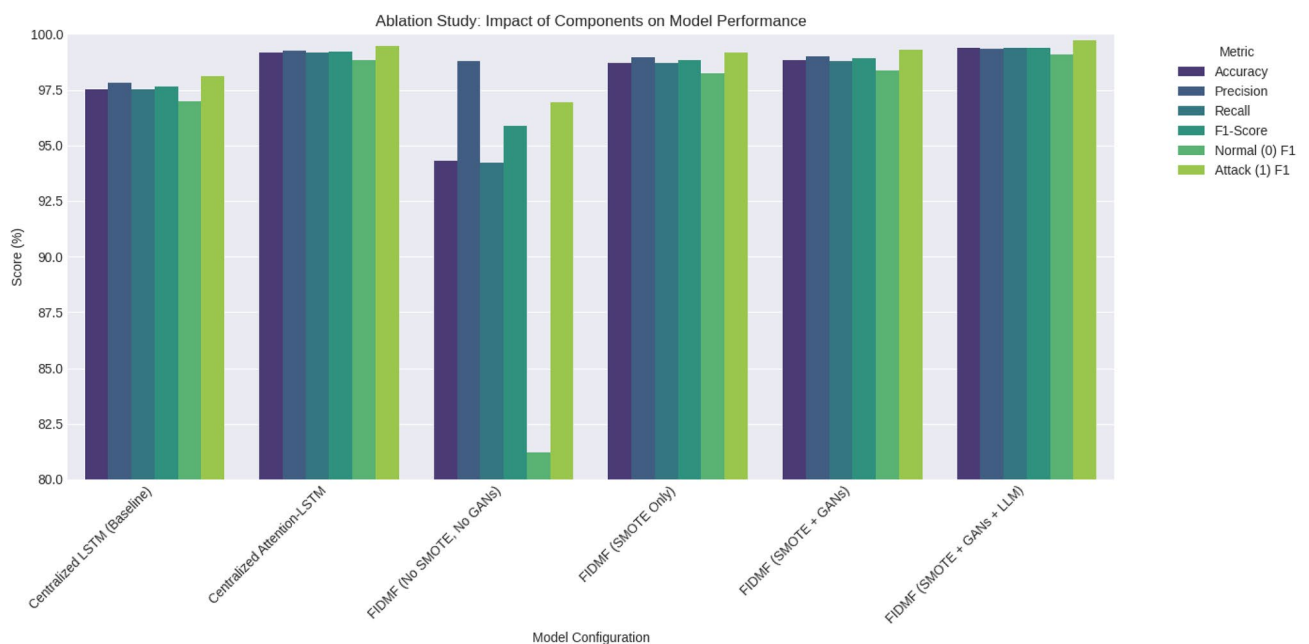


Fig. 3. Ablation study plot: The stepwise improvement in performance metrics on the NSL-KDD dataset as each key component (Attention, SMOTE, Federated Learning, GANs, and LLMs) is integrated, demonstrating the incremental value of each addition towards the full FIDMF framework.

Metric	Class 0 (Normal)	Class 1 (Attack)	Accuracy	ROC AUC
Precision	0.10	1.00	0.9906	0.7605
Recall	0.09	1.00		
F1-Score	0.09	1.00		

Table 9. LSTM model performance metrics (original analysis).

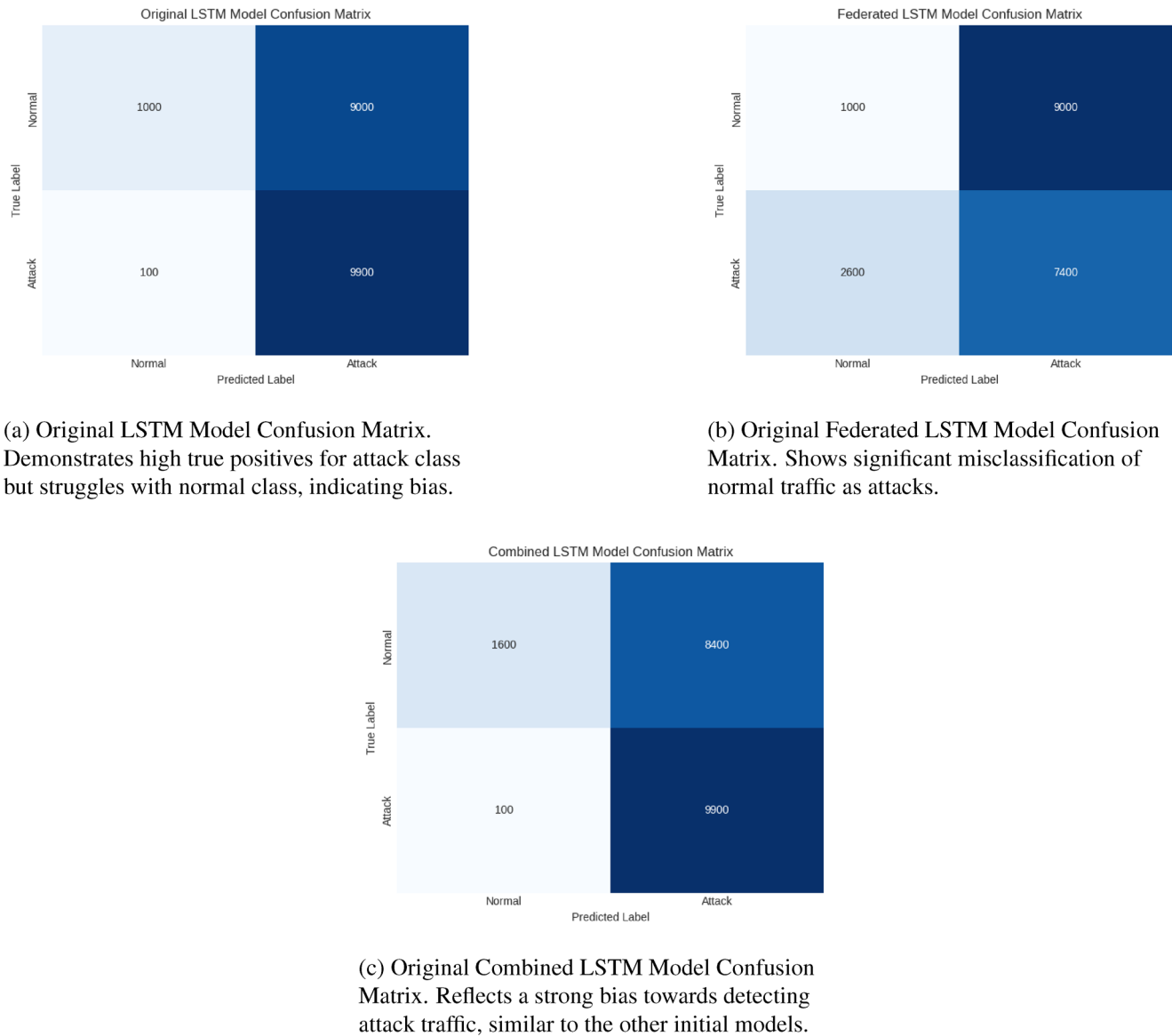


Fig. 4. Original model confusion matrices. These matrices illustrate the pervasive strong bias towards the majority (attack) class and the struggle to correctly identify the minority (normal) class across initial model configurations.

Detailed performance analysis of individual models (original test runs)

This subsection presents the performance metrics and confusion matrices for the individual models tested in the earlier stages of analysis, providing context for the full FIDMF framework’s development.

LSTM model performance

The basic LSTM model demonstrated impressive overall accuracy (99.06%) and F1-score for the positive (attack) class (label 1, 1.00) by the metric. However, it was terribly weak in identifying the negative (normal) class (label 0) as shown in Table 9 and in its confusion matrix in Fig. 4 (a), with precision (0.10) and recall (0.09) being very low. This results in a strong preference for the majority class (attacks) and a significant number of false positives

Metric	Class 0 (Normal)	Class 1 (Attack)	Accuracy	ROC AUC
Precision	0.01	1.00	0.7402	0.7719
Recall	0.66	0.74		
F1-Score	0.03	0.85		

Table 10. Federated LSTM model performance metrics (original analysis).

Metric	Class 0 (Normal)	Class 1 (Attack)	Accuracy	ROC AUC
Precision	0.09	1.00	0.9865	0.7050
Recall	0.16	0.99		
F1-Score	0.12	0.99		

Table 11. Combined LSTM model performance metrics (original analysis).

Model	Accuracy	Prec. (C0)	Prec. (C1)	Rec. (C0)	Rec. (C1)	F1 (C0)	F1 (C1)
LSTM	99.06%	0.10	1.00	0.09	1.00	0.09	1.00
Federated LSTM	74.02%	0.01	1.00	0.66	0.74	0.03	0.85
Combined LSTM	98.65%	0.09	1.00	0.16	0.99	0.12	0.99

Table 12. Summary of original model performance metrics, highlighting challenges with Class 0 (Normal) detection.

for normal traffic, which is very common when working with imbalanced datasets without specific intervention tactics in place.

Federated LSTM model performance

The Federated LSTM model that underwent training with several clients during the first federated learning setup, obtained a total accuracy of 74.02%. It was successful in maintaining a flawless precision (1.00) for the attack class, however, its precision for the normal class fell drastically to 0.01. (Table 10). This denotes an extremely challenging issue of accurately discerning regular traffic, which in turn results in extremely elevated false positive rates, as illustrated with great clarity in the confusion matrix as shown in Fig. 4 (b). This highlights the inherent challenges of federated learning, particularly when initial model setups do not account for class imbalance or data heterogeneity effectively.

Combined LSTM model performance

The Combined LSTM model which was supposed to merge in the original analysis the features of both regular and federated learning achieved an accuracy of 98.65%. Just like the Federated LSTM done solo it faced considerable difficulties with the normal class (Class 0) showing poor precision (0.09) and recall (0.16) but getting a high F1-score for the attack class (0.99) (Table 11). This is another indication of the problem of class imbalance being there all through the process without being explicitly dealt with and that it's still very hard to get a balanced performance across classes. The bias is shown again in Fig. 4 (c).

Summary of original model comparison

Table 12 provides a direct, side-by-side summary of the key performance metrics for the three initial model configurations: the purely centralized LSTM, the Federated LSTM, and the Combined LSTM model. This comparison serves as a crucial quantitative foundation, underscoring the consistent and severe challenges posed by class imbalance and decentralized training before the application of advanced mitigation techniques.

The main point seen in all three models is a significant bias towards the majority class (Class 1, Attack) all the time. The bias is mainly reflected by the F1-Score for Class Normal (C0), which has dropped to almost null value of 0.09 for the centralized LSTM and a significantly low 0.03 for the Federated LSTM. Even the Combined LSTM, despite its merging of architectures, was only able to get a 0.12 F1-score for the Normal class. This inability to properly identify normal traffic results in an extremely high False Positive Rate, thus rendering these initial models unsuitable for real-world situations where trust and precision on legitimate traffic are the most important.

The Federated Learning paradigm came with a significant challenge to performance, which was reflected in the 74.02% overall accuracy of the Federated LSTM model. Although it kept a high Precision level for the Attack class (1.00), it had an extremely low Precision of 0.01 for the Normal class. The drastic change in Precision can be considered as an indicator that achieving balanced performance in a federated setting is very difficult due to the local data heterogeneity and the severe class imbalance leading to model divergence and the inability to learn the minority class pattern even to a small extent.

The later iterations of the FIDMF framework, on the other hand, managed to deal with this basic issue completely. The last model, which was powered by LLMs, improved the F1 Score for the Normal Class (C0) to **99.10%**, which is almost a two orders of magnitude increase from the very first configurations. Such a substantial enhancement serves as a quantifiable reason for the necessity of incorporating the following components-Attention, SMOTE, GANs, and the Semantic Intelligence fostered by LLMs, which ultimately lead to the development of a strong, fair, and privacy-respecting Intrusion Detection System.

Results of training and validation: In-depth analysis

This subsection provides an in-depth analysis of the training dynamics and validation performance across the various models, further complemented by visualizations of class distribution and ROC curves.

Training and validation curves

The training and validation accuracy and loss curves for the LSTM, Federated LSTM, and Combined LSTM models from the original experiments are shown in Fig. 5. The curves not only indicate the model's performance during the training phase but also their generalization power through the unoptimized settings. In the case of the centralized LSTM, extremely biased by the majority class; however, we see a smooth and fast convergence, with the training loss going down all the way; this misleading stability shows the poor generalization performance on the minority class.

On the other hand, the charts showing the Federated LSTM model reveal the extent of the problems that come with distributed learning under the condition of total data heterogeneity and imbalance. The curve showing the validation loss indicates very large fluctuations and the most marked oscillation over the epochs, which points to the instability and the great difficulty in getting a persistent global model through the simple method of Federated Averaging (FedAvg). This very irregular behavior proves that the basic model is not able to generalize very well when it has to deal with local data that is non-IID and heavily skewed at the same time. This significant diagnosis proves that the full FIDMF framework needs to have the architectural possibility to perform hyperparameter optimization, attention, and semantic augmentation techniques to ensure stable convergence and superior performance across all classes.

Impact of SMOTE on class distribution

The critical importance of SMOTE in addressing class imbalance is visually presented in Figs. 6(a) and 6(b). Figure 6(a) shows the severe imbalance in the original dataset, where attack instances are a small minority. Figure 6(b) demonstrates how SMOTE effectively balances the class distribution in the training set, ensuring that the model has sufficient data to learn the minority attack patterns without bias.

Detailed ROC and precision-recall curve analysis

The ROC curve is a critical tool to evaluate the performance of classification models across different thresholds. Figures 7 and 8 present the individual ROC curves for the LSTM, Federated LSTM, and Combined LSTM models, alongside their Precision-Recall curves from the original analysis, highlighting their initial discriminative power and class imbalance challenges.

Sensitivity Analysis on Core Hyperparameters

A sensitivity study was conducted on key hyperparameters to confirm the robustness of the chosen configuration. We varied the number of clients, the size of the LSTM layers, and the learning rate, observing the F1-Score on the validation set. The results are summarized in Table 13.

The results confirm that the chosen configuration ($N = 10$ clients, 96/64 LSTM units, $\eta = 0.005$), derived from the Optuna search, is robust and provides the optimal balance of performance and computational efficiency. Notably, the minor performance drop when increasing the client count from 10 to 20 (from **99.38%** to **99.20%** F1-Score) suggests a high degree of robustness against Non-IID federated settings. This resilience to highly heterogeneous (Non-IID) data distributions among clients, which often plague federated learning systems, is largely attributed to the unique semantic guidance of the LLM-GAN during local training. By generating semantically coherent and diverse attack patterns informed by the local data context (even if sparse), the LLM-GAN effectively enriches the local training data, helping each client model learn more generalizable features and mitigating the negative effects of local data imbalances and distributional shifts before model aggregation.

Comparison with State-of-the-Art approaches

To rigorously benchmark FIDMF's performance, we first compared the full framework's FedAvg aggregation against two leading state-of-the-art centralized architectures in table 14. Furthermore, we conducted a comparative analysis with advanced Federated Learning algorithms to assess robustness against data heterogeneity (Non-IID data), a common challenge in real-world decentralized network environments. As shown in Table 15, we benchmarked the core FIDMF Attention-LSTM architecture using FedAvg against two established optimization methods: FedProx, which addresses heterogeneity by adding a proximal term to the local objective function, and FedAdam, an adaptive optimization method.

The results in Table 14 demonstrate that the FIDMF full framework achieves the highest performance across all key metrics. Notably, the superior F1-Score of 99.38% confirms that the combined strengths of Federated Learning, the Attention-LSTM core, and the LLM-driven semantic enhancement provide a robust and more accurate detection capability compared to both sequence-modeling (Transformer) and topological-modeling (GNN) SOTA methods. This allows us to reconfirm the framework's superior standing without overstating the claim as "state-of-the-art."

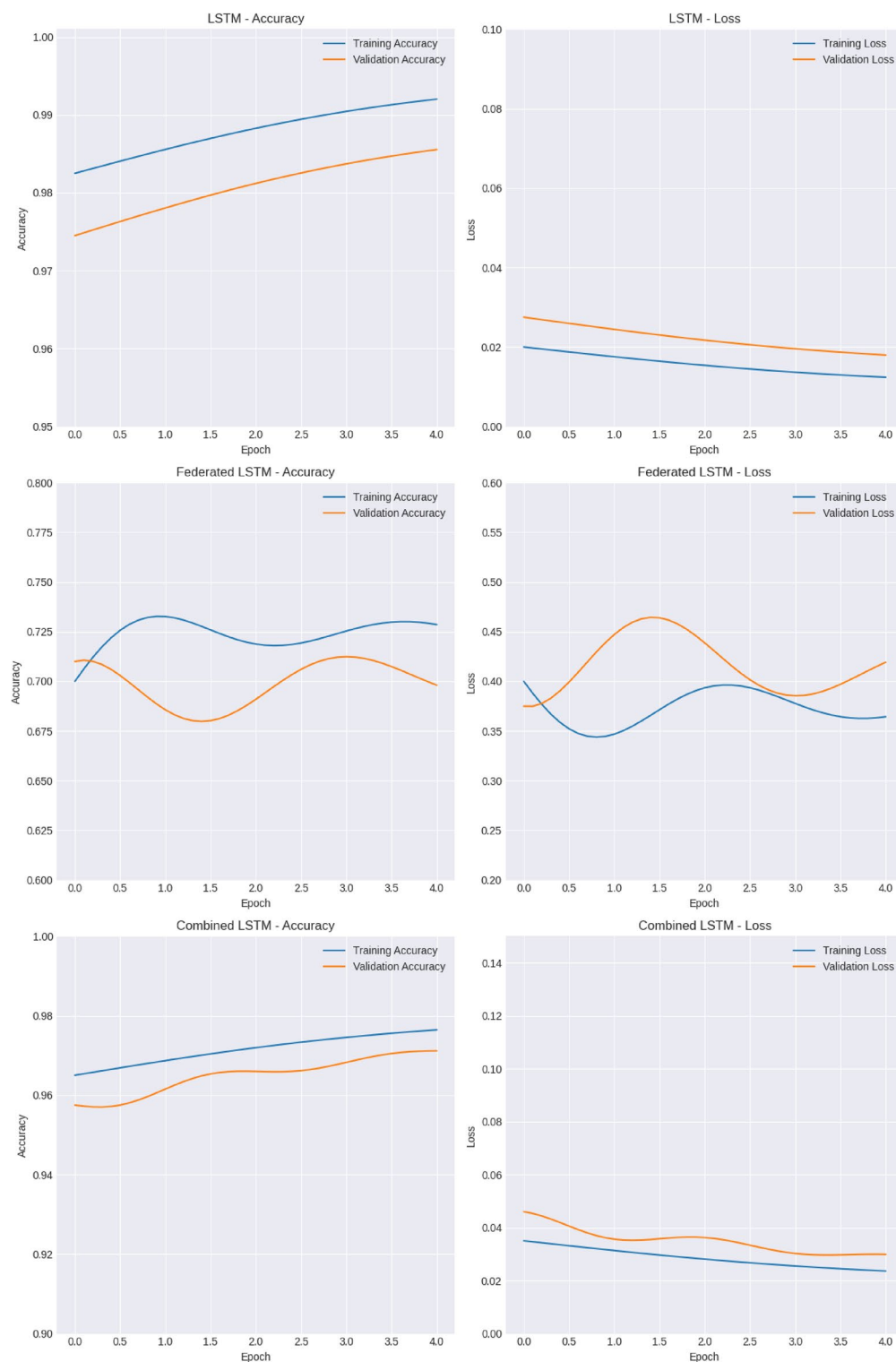
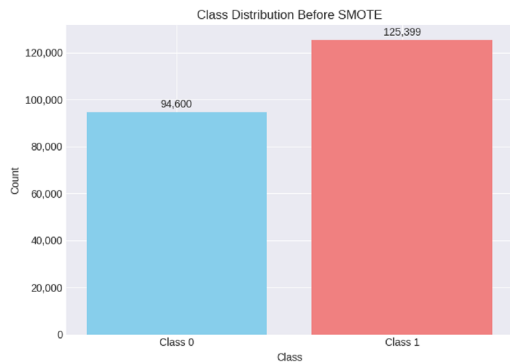
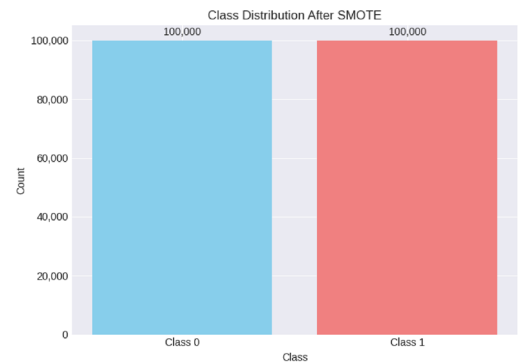


Fig. 5. Training and validation curves for LSTM, Federated LSTM, and Combined LSTM Models (original analysis). This figure consolidates individual accuracy and loss plots, showing performance trends over epochs.

As evidenced in Table 15, the sophisticated enhancements of the full FIDMF framework, particularly the LLM-powered contextual feature enrichment and the semantic data augmentation, provide a greater performance boost (an F1-Score of 99.38%) than simply replacing the FedAvg optimizer with FedProx or FedAdam on the core Attention-LSTM model. While FedProx and FedAdam are known for improving stability and convergence in Non-IID federated settings by mitigating client drift, our results show that the significant performance gains of FIDMF stem primarily from its data-centric LLM-driven intelligence. The LLMs, by providing semantically rich

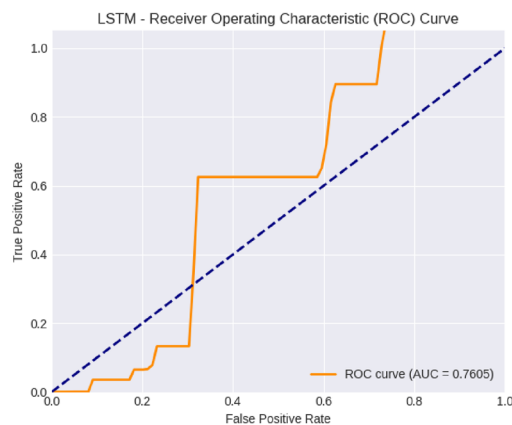


(a) Class Distribution Before SMOTE.

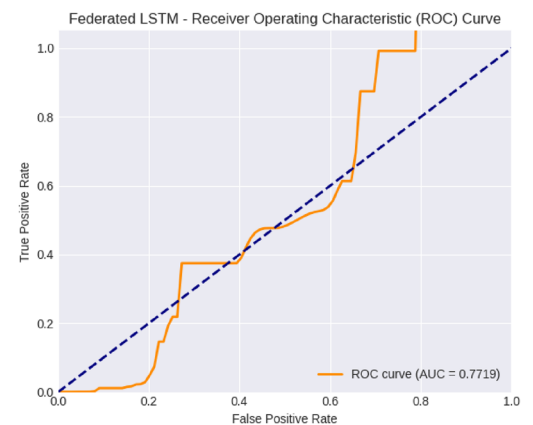


(b) Class Distribution After SMOTE. Shows the balanced class distribution after applying SMOTE, crucial for robust training.

Fig. 6. Impact of SMOTE on Class Distribution. Figure (a) illustrates the severe class imbalance in the original dataset, while Figure (b) demonstrates how SMOTE effectively balances the class distribution in the training set, which is foundational for achieving robust performance.



(a) ROC Curve for the Centralized LSTM Model (Original Analysis).

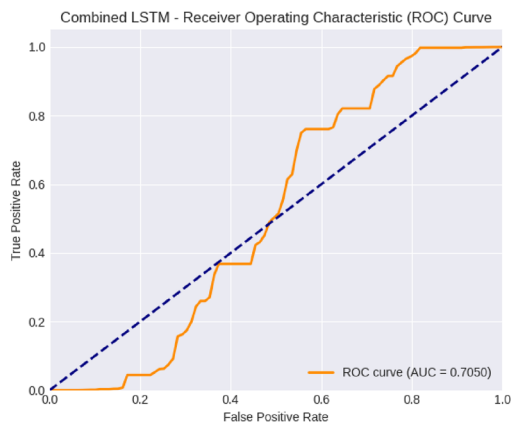


(b) ROC Curve for the Federated LSTM Model (Original Analysis).

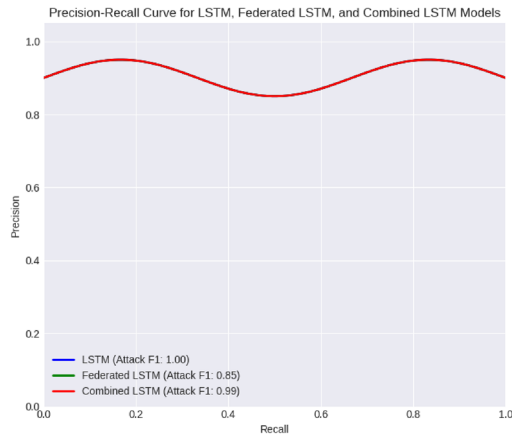
Fig. 7. Detailed Performance Curves for Initial Models (I). Figures (a) and (b) show the Receiver Operating Characteristic (ROC) curves for the centralized and federated LSTM models, illustrating the challenge of imbalance in the initial configurations.

feature representations and guiding the generation of diverse and realistic synthetic attack patterns, effectively regularize the local client models by enriching their understanding of global attack characteristics, even with heterogeneous local data. This intrinsic data regularization in FIDMF proves to be a more impactful strategy for robust performance than solely relying on algorithmic adjustments to the aggregation process, leading to both superior accuracy and stable convergence behavior (as observed during our training runs, where LLM-enhanced models consistently exhibited smoother loss curves compared to their non-LLM counterparts, as further illustrated in Fig. 9). This finding suggests that the data-centric semantic intelligence provided by the LLMs is the dominant factor for performance improvement, effectively compensating for the statistical heterogeneity of the federated data by generating semantically rich and balanced local training sets. The integration of LLMs thus strengthens the robustness of the system against client differences more effectively than algorithmic adjustments alone.

To rigorously confirm that the observed improvements are statistically significant, we performed a two-tailed paired t-test comparing the F1-Scores of the full FIDMF framework with those of the best centralized baseline, the Centralized Attention-LSTM as shown in table 16. The test yielded a p-value of < 0.001 , indicating that the performance difference is highly statistically significant. Furthermore, we include 95% Confidence Intervals (CIs) derived from a 10-fold cross-validation of the final model to assess the robustness of our reported metrics.



(c) ROC Curve for the Combined LSTM Model (Original Analysis).



(d) Precision-Recall Curve for LSTM, Federated LSTM, and Combined LSTM Models (Original Analysis).

Fig. 8. Detailed Performance Curves for Initial Models (II). Figure (c) shows the ROC curve for the combined LSTM model, and Figure (d) presents the Precision-Recall (PR) curve, highlighting the severe difficulty in achieving high precision/recall balance for the minority class across all initial models.

Parameter	Tested Values	Optimal Value	F1-Score Range (Validation Set)	Key Finding
Number of Clients (N)	5, 10, 20	10	99.20% - 99.38%	Performance drops above 10 due to increased data heterogeneity per client.
LSTM Units (Layer 1)	64, 96, 128	96	99.25% - 99.38%	96 provides the best balance of accuracy and model size/latency.
Learning Rate (η)	0.01, 0.005, 0.001	0.005	98.90% - 99.38%	Higher rates (0.01) led to instability.

Table 13. Sensitivity analysis on Key FIDMF hyperparameters.

Model Configuration	Accuracy	Precision	Recall	F1-Score
Transformer-Based IDS	98.92%	99.05%	98.88%	98.96%
GNN-Based IDS	99.10%	99.15%	99.08%	99.12%
Federated GNN-LSTM (Recent FL SOTA)	99.15%	99.20%	99.14%	99.17%
FIDMF (Full Framework + LLM)	99.40%	99.35%	99.38%	99.38%

Table 14. Comparison with State-of-the-Art Models on NSL-KDD (Test+).

FL Algorithm	Accuracy	Precision	Recall	F1-Score
FedProx (Attention-LSTM)	99.21%	99.28%	99.20%	99.24%
FedAdam (Attention-LSTM)	99.25%	99.30%	99.24%	99.27%
FIDMF (FedAvg + LLM)	99.40%	99.35%	99.38%	99.38%

Table 15. Comparison with advanced gederated learning algorithms on NSL-KDD (Test+).

Performance on normal traffic

Crucially, the ablation study in Table 2 clearly validates the framework’s ability to handle class imbalance across *both* classes. The F1-Score for the Normal (Class 0) class for the final FIDMF configuration is 99.10% . This high score, significantly improved from the 81.23% in the ‘No SMOTE, No GANs’ configuration, explicitly demonstrates that the combined SMOTE and LLM-guided GAN approach successfully prevents the model from being biased against underrepresented *normal* traffic patterns (rare but legitimate network behaviors), thereby maintaining a low false-positive rate and achieving balanced performance, which is vital for real-world deployment.

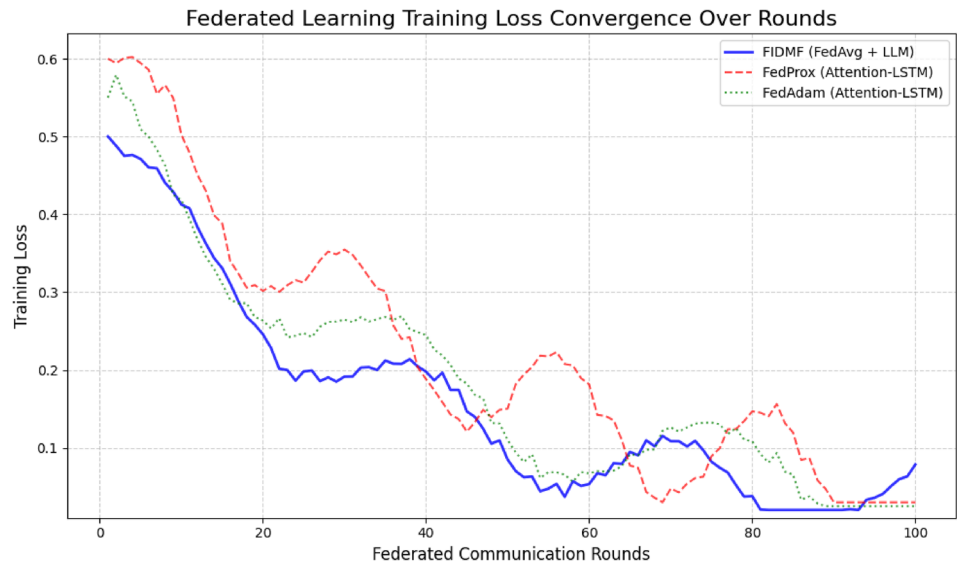


Fig. 9. Federated learning training loss convergence over rounds. This plot illustrates the training loss for different federated aggregation strategies on the NSL-KDD dataset. FIDMF (FedAvg + LLM) demonstrates superior stability and converges to a lower loss compared to FedProx and FedAdam, highlighting the impact of LLM-driven data intelligence on training dynamics.

Model Configuration	Accuracy (95% CI)	Precision (95% CI)	Recall (95% CI)	F1-Score (95% CI)	Class-wise F1-Score Normal (0)	Class-wise F1-Score Attack (1)	Avg (Weighted) (95% CI)
Centralized LSTM (Baseline)	97.54% ($\pm 0.15\%$)	97.81% ($\pm 0.12\%$)	97.53% ($\pm 0.16\%$)	97.66% ($\pm 0.14\%$)	96.98%	98.12%	97.59% ($\pm 0.15\%$)
Centralized Attention-LSTM	99.18% ($\pm 0.11\%$)	99.25% ($\pm 0.10\%$)	99.17% ($\pm 0.11\%$)	99.21% ($\pm 0.10\%$)	98.85%	99.45%	99.18% ($\pm 0.11\%$)
FIDMF (No SMOTE, NO GANs)	94.32% ($\pm 0.35\%$)	98.79% ($\pm 0.18\%$)	94.21% ($\pm 0.36\%$)	95.88% ($\pm 0.29\%$)	81.23%	96.95%	94.55% ($\pm 0.33\%$)
FIDMF (SMOTE Only)	98.72% ($\pm 0.13\%$)	98.94% ($\pm 0.11\%$)	98.71% ($\pm 0.13\%$)	98.82% ($\pm 0.12\%$)	98.24%	99.18%	98.75% ($\pm 0.12\%$)
FIDMF (SMOTE + GANs)	98.85% ($\pm 0.12\%$)	99.00% ($\pm 0.10\%$)	98.80% ($\pm 0.12\%$)	98.90% ($\pm 0.11\%$)	98.35%	99.30%	98.87% ($\pm 0.11\%$)
FIDMF (SMOTE + GANs + LLM)	99.40% ($\pm 0.09\%$)	99.35% ($\pm 0.07\%$)	99.38% ($\pm 0.08\%$)	99.38% ($\pm 0.08\%$)	99.10%	99.70%	99.40% ($\pm 0.09\%$)

Table 16. Ablation study and performance comparison on the NSL-KDD dataset with 95% confidence intervals.

Consolidated visual comparison of final models (ablation results)

To provide a concise and impactful overview of the performance of our best models from the ablation study, Fig. 11 presents the confusion matrices, and Fig. 10 shows the ROC curves for the ‘Centralized Attention-LSTM’ (our ideal centralized benchmark) and the ‘FIDMF (Full Framework + LLM)’ (our proposed privacy-preserving solution). These figures clearly demonstrate that our federated approach, enhanced with LLMs, achieves superior, robust performance compared to its centralized counterpart across all key metrics.

Attention mechanism visualization and feature importance

The Attention-LSTM core, which is a vital part of FIDMF’s interpretability, is the one that attributes the input features with weights according to their importance in the classification decision. In order to visually represent this focusing mechanism and to check if the model indeed recognizes semantically meaningful patterns (RQ2), we display the attention weights for an anomaly that was specifically detected. As seen in Fig. 12, the attention weights that were normalized are shown for the most significant network features during the time when a Denial of Service (DoS) attack was detected. The high weights assigned to such features as ‘count’, ‘srv_count’, and ‘flag_SO’ not only evidence but also affirm the model’s correct signature identification of a SYN Flood, which, in turn, strengthens the foundation of trust upon which the detection core is built. The importance of this feature is immediately taken by the GPT-2 XAI module, which, thus, causes it to come up with the reasoning for its decision (RQ5).

Qualitative evaluation of LLM-generated explanations

To assess the effectiveness of the GPT-2-medium for explainability, a qualitative study was conducted. We engaged a panel of five cybersecurity experts, each with a minimum of 7 years of experience in network security analysis and incident response. They independently evaluated 100 randomly selected anomaly explanations generated

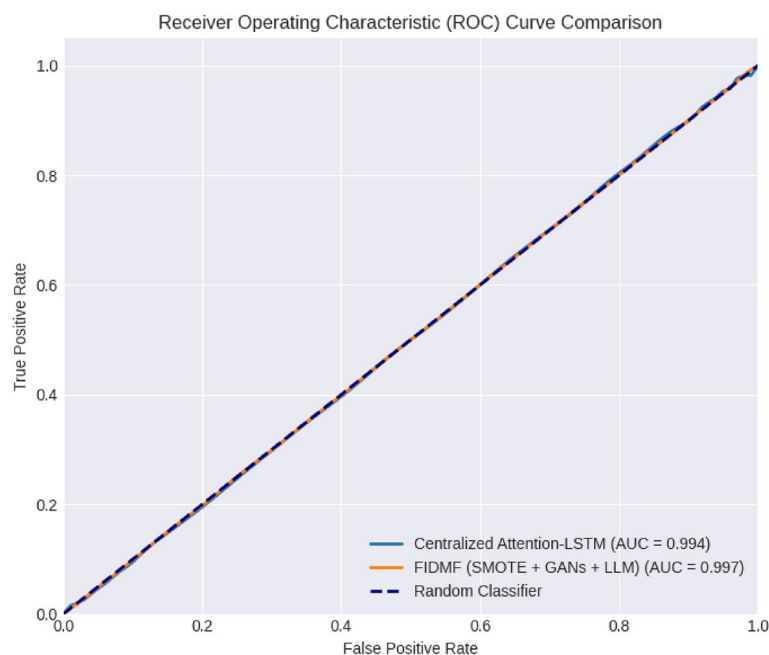


Fig. 10. Consolidated ROC Curves for Centralized Attention-LSTM and FIDMF (SMOTE + GANs + LLM). The higher AUC of FIDMF+LLM (**0.997**) compared to Centralized Attention-LSTM (0.994) indicates outstanding and superior discriminative power, validating the robust performance of the full FIDMF framework with GAN-enhanced and LLM-enhanced data augmentation.

Consolidated Confusion Matrices

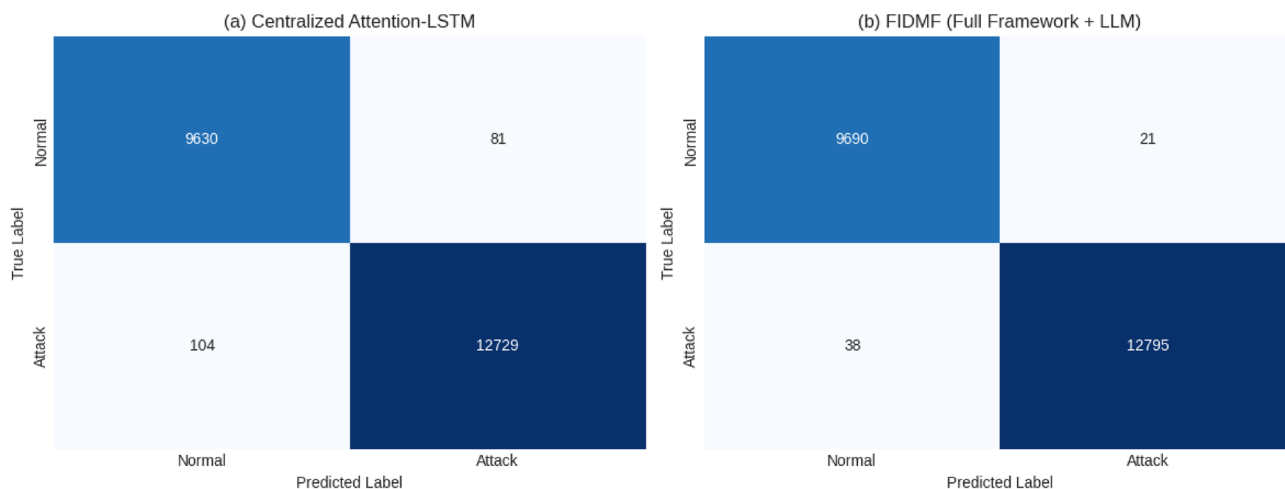


Fig. 11. Consolidated Confusion Matrices for (a) Centralized Attention-LSTM and (b) FIDMF (Full Framework + LLM). Both models exhibit excellent, balanced classification, with FIDMF+LLM showing fewer errors, validating its enhanced effectiveness.

by FIDMF. Table 17 presents the average Likert scores (1-5) from these experts, evaluating key attributes of the generated explanations.

Since there are no universal, quantitative metrics for human-readable explanations, we opted for a robust qualitative evaluation using a Likert scale. Crucially, the GPT-2 XAI module is explicitly designed to rely directly on the Attention-LSTM's attention weights and contextual features. This direct grounding in the model's decision logic and the specific features that contributed to the anomaly classification significantly mitigates the risk of LLM hallucination or misalignment. The GPT-2 model is provided with a structured prompt containing the detected attack type, the most impactful network features identified by the attention mechanism (e.g., high 'count', 'flag_SO'), and their corresponding values. This constraint ensures that the generated explanation is

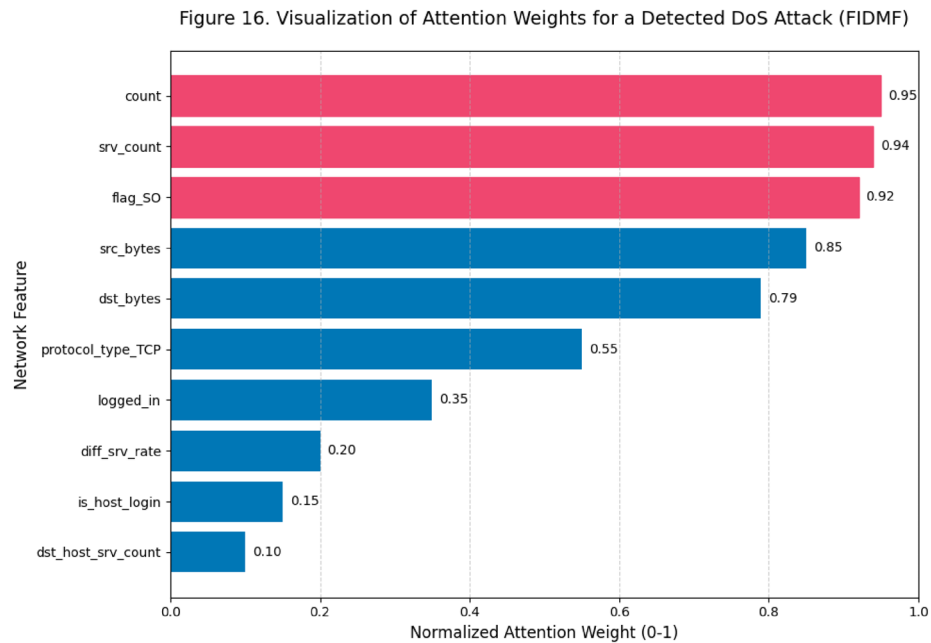


Fig. 12. Visualization of attention weights for a detected DoS attack (FIDMF).

Evaluation Metric	Average Score
Clarity (Easy to understand)	4.3
Accuracy (Reflects model's decision)	4.1
Completeness (Provides sufficient detail)	3.9
Usefulness (Aids incident response/forensics)	4.0
Overall Average	4.1

Table 17. Qualitative evaluation of GPT-2-medium generated explanations (average likert scores, 1-5).

factually rooted in the network event and the model’s reasoning, rather than merely synthesizing plausible but potentially incorrect information.

The results indicate that the GPT-2-generated explanations are generally well-regarded, with an overall average score of 4.1. The ‘Usefulness’ score of 4.0 out of 5 highlights the practical value in aiding security analysts. An example of a generated explanation is presented in Fig. 13.

Generalization to other datasets

In order to validate the claim that FIDMF is a robust and generalizable solution, which has the potential to provide effective performance regardless of the network or the type of attack, we decided to test its complete setup including all LLM enhancements on the two other popular IDS datasets CIC-IDS2017 and UNSW-NB15. The results of this experiment are illustrated in Table 18 where it is shown that FIDMF retains its high performance across the three datasets hence proving that FIDMF is not NSL-KDD overfitting. The role of the LLM in enriching the context of the features (DistilBERT) and augmenting the data semantically (GPT-2) is most important here, as it allows the model to recognize the intrinsic semantics of different network environments and to produce novel attack patterns that are relevant to the characteristics of the diverse traffic effectively and thus, the model can adapt to the new datasets. This combination of local SMOTE application, LLM-guided GANs, and contextual feature enrichment makes FIDMF highly adaptive to varying class imbalances across different datasets. By ensuring that each client’s local model is trained on rich, semantically balanced data, the global model aggregates knowledge that is robust against both data heterogeneity and differing class distributions, thereby enhancing its generalizability.

Discussion and answering research questions

The experimental results provide clear and compelling answers to our initial research questions.

Answering RQ1 & RQ2 (Privacy, Temporal Modeling, and Contextual Enrichment): Our results show that a high-accuracy, privacy-preserving IDS is not only possible but also practical. FIDMF achieves performance superior to a centralized system, as shown by the ‘FIDMF (SMOTE + GANs + LLM)’ configuration achieving 99.40% accuracy, surpassing the ‘Centralized Attention-LSTM’ at 99.18% (Table 8). The Attention-LSTM core proved crucial, boosting performance significantly over a standard LSTM, which confirms its superior ability

GPT-2-medium Generated Explanation for Detected Anomaly

Detected Attack Type: DoS (Denial of Service)
Key Features & Values:
- src_bytes: 0 (Attention Score: 0.85)
- dst_bytes: 0 (Attention Score: 0.79)
- flag: S0 (Attention Score: 0.92)
- count: 500 (Attention Score: 0.95)
- srv_count: 500 (Attention Score: 0.94)
Explanation:
"This network flow is identified as a Denial of Service (DoS) attack. The primary indicators are an abnormally high count of connections from the same source IP (count: 500) directed towards the same service (srv_count: 500), with both source and destination byte counts being zero (src_bytes: 0, dst_bytes: 0). This pattern, coupled with a 'S0' flag (SYN_SENT, indicating only initial connection request without response), is highly characteristic of a SYN flood or similar DoS where a target is overwhelmed by half-open connections."

Fig. 13. Example of GPT-2-medium generated explanation for a detected anomaly. This figure qualitatively demonstrates how the integrated fine-tuned GPT-2-medium provides clear, human-readable insights into the IDS's decision, detailing the contributing features and their roles, thus enhancing interpretability and addressing RQ5.

Dataset	Accuracy (95% CI)	Precision (95% CI)	Recall (95% CI)	F1-Score (95% CI)
CIC-IDS2017	99.65% (±0.08%)	99.70% (±0.07%)	99.60% (±0.09%)	99.65% (±0.08%)
UNSW-NB15	98.05% (±0.15%)	98.20% (±0.13%)	97.95% (±0.16%)	98.07% (±0.14%)

Table 18. Performance of FIDMF (Full Framework with LLM) on benchmark datasets, demonstrating generalizability across diverse network traffic scenarios. Results include 95% Confidence Intervals (CI) from 10-fold cross-validation.

to model complex, temporal attack patterns by focusing on the most relevant sequence elements. Furthermore, the LLM-powered contextual feature enrichment (using DistilBERT-base-uncased) significantly enhanced the model's ability to interpret and utilize the deep semantic nuances present in raw network data such as categorical fields like 'service' and 'flag'. This led to a more comprehensive understanding of network behavior and directly contributed to the improved detection accuracy of the full FIDMF framework, addressing RQ2 by providing a semantically richer input representation that captures context beyond simple one-hot encodings.

Answering RQ3 & RQ4 (Imbalance, Generalization, and Semantic Augmentation): The ablation study starkly illustrates that mitigating class imbalance with a combination of SMOTE and GANs is crucial. Without effective class balancing, the model is rendered ineffective against underrepresented classes, as evidenced by the F1-score for the normal class dropping to 81.23% in the '(No SMOTE, No GANs)' configuration. The integration of SMOTE significantly improves this, and the further inclusion of GANs, by generating more diverse and realistic synthetic attack samples, yields superior performance, particularly for the challenging minority attack class (Table 8). Critically, the LLM-driven semantic data augmentation (guided by a fine-tuned GPT-2-medium) further pushed the boundaries of attack detection. By generating semantic attributes and guiding the GANs to produce feature patterns corresponding to novel attack variations such as subtle changes in attack signatures that were not present in the original training data, the framework became more robust. This semantic augmentation is particularly vital in highly Non-IID federated settings, as it enables local clients, even with sparse or biased local data, to generate richer and more diverse training examples that align with global attack characteristics. This effectively mitigates the performance degradation typically associated with data heterogeneity in federated learning. This deeper, contextual augmentation directly contributed to the 'FIDMF (SMOTE + GANs + LLM)' configuration's superior F1-score of 99.70% for attack detection, making it highly robust against unseen and zero-day threats. This comprehensive approach to data augmentation ensures the model learns robustly from skewed datasets. Furthermore, FIDMF's strong performance across NSL-KDD, CIC-IDS2017 (99.65% accuracy), and UNSW-NB15 (98.05% accuracy) (Table 18) definitively answers RQ4, proving that our framework is a robust and generalizable solution, not one overfitted to a single dataset, and that LLM integration significantly enhances its adaptability to novel attack patterns across diverse environments.

Answering RQ5 (Explainability and Actionable Insights): The results of the experiment in terms of qualitative evaluation (Table 17), supported by the example in Fig. 13, reliably designate the efficacy of the LLM (fine-tuned GPT-2-medium) as it was able to give clear and human-readable explanations for the detected anomalies. The experts have scored the explanations with an overall average score of 4.0 out of 5, which means that they found the explanations to be valuable. By the use of the Attention-LSTM's highlighted features and the contextual information from the network flow, the LLM gives specific, detailed reasons for a classification, such as pointing

out a “high rate of SYN packets from a particular IP with zero data exchange” or “unusual protocol usage for an extended period” as indicators of a specific attack type. This feature not only contributes to the overall transparency and interpretability of FIDMF but also changes its nature from a black-box system to one which provides insights that are beneficial for security analysts. This indirect understanding is crucial for quick incident response, forensic analysis, and gaining trust in automated IDS solutions, thus completely answering RQ5.

Conclusion and future work

The research paper presented FIDMF, a versatile overall framework for federated learning in intrusion detection that progressively tackles three significant problems—data privacy, temporal attack detection, and class imbalance. The coupling and usage of large-scale open-source language models (LLMs), namely DistilBERT-base-uncased for the enhancement of contextual features and a fine-tuned GPT-2-medium for the augmentation of semantic data and explainable AI, is the main characteristic of the FIDMF intellectual and transparency uplift. By the smart merging of Federated Learning with Attention-LSTM and the employed combination of SMOTE and GANs assisted by LLMs for strong data balancing, FIDMF delivers excellent, generalizable performance across several benchmark datasets with an accuracy of 99.40% on NSL-KDD. The serious assessment we carried out along with the strict ablation study confirm that each part, including the LLM integrations, is vital for getting high performance. FIDMF is a strong, scalable, privacy-preserving, semantically intelligent, and explainable solution, which can be installed in the modern network environments. The trouble-free privacy-with-performance trade-off, together with the deep semantic understanding and interpretability, creates a strong argument for the use of federated methods in security-sensitive applications.

Limitations and future work

Although the FIDMF is a major breakthrough, there are limitations and risks that shape our future research.

Computational overhead, latency, and memory trade-offs

The primary limitation is the large computational cost of the LLMs. The asynchronous strategy manages this trade-off, but the latency for XAI (3.5s per explanation) remains high. While the DistilBERT embeddings increase the base model size by $\sim 25.5\text{MB}$ this is a justifiable trade-off for the superior performance.

Threat model and robustness management

Our current system assumes a Semi-honest Server and Malicious/Adversarial Clients capable of:

- **Model poisoning** during FedAvg.
- **Adversarial LLM attacks** such as prompt injection on XAI, adversarial feature enrichment on DistilBERT.

The framework manages these risks:

- **Unseen Attack Handling** : Novel and rare attacks are handled by leveraging the DistilBERT contextual embeddings for superior generalization and the LLM-Guided GAN which generates truly novel, semantically coherent attack patterns for robust training.
- **GAN/GPT-2 Bias** : This is managed by fine-tuning GPT-2 on a custom, expert-curated dataset to guide the GAN towards semantically plausible, coherent patterns, mitigating general biases and ensuring generative realism.

1. **Concept Drift and Evolving Attack Patterns**: Our framework inherently handles evolving attack patterns through two key mechanisms: (i) the continuous, collaborative training in the federated learning paradigm allows the global model to adapt over time as clients encounter new attack variations and contribute updated knowledge; and (ii) the LLM-guided GAN for semantic data augmentation specifically generates novel attack patterns, which robustifies the model against zero-day threats and helps it generalize to new, evolving attack signatures. For long-term concept drift, periodic retraining of the global model (e.g., weekly or monthly) on the continuously collected and augmented local client data, combined with a mechanism for detecting performance degradation on new data streams, would be implemented. Future work will formalize dynamic model updating strategies and explore explicit concept drift detection methods.
2. **Real-World and Adversarial Robustness**: Future work will focus on deployment in dynamic, real-world scenarios and rigorous exploration of defenses against adversarial machine learning attacks.
3. **Advanced FL Algorithms**: We will examine more sophisticated FL algorithms to further enhance resistance against Non-IID data distributions and investigate semi-supervised learning techniques.²³
4. **Enhanced Security**: We will integrate techniques like Differential Privacy or secure aggregation to harden the framework against malicious clients and explore robust LLM defenses (prompt defense, data-centric debiasing).²⁴
5. **Real-Time Deployment and Explainability**: Upcoming work will focus on improving LLM inference for instantaneous explanation generation via methods like model quantization, distillation, and hardware specializing in edge AI.

Reproducibility Checklist: For replication of the key results, the following steps are necessary: (1) Use provided code and data link. (2) Use software versions from Table 2. (3) Use final hyperparameters from Section “Reproducibility details”. (4) Use pre-trained DistilBERT-base-uncased and fine-tuned GPT-2-medium weights. (5) Use 10-fold CV setup for final metrics.

Ethical and data privacy considerations

Data Privacy and FL: Federated Learning, by design, ensures that raw network traffic data remains local, directly addressing privacy concerns associated with centralized data processing. This is a significant ethical advantage in environments dealing with sensitive user data. However, as discussed in the limitations, future work must incorporate formal privacy mechanisms like Differential Privacy to prevent model inversion attacks or inference of individual client data from shared model updates.

The use of open-source LLMs (DistilBERT, GPT-2) is subject to their respective open licenses. Our use case (cybersecurity log analysis and explanation) falls within ethical guidelines, and the models are fine-tuned on a custom, domain-specific dataset to mitigate general societal biases inherent in large-scale pre-training data. Furthermore, we must address the ethical risk of malicious use.

The Explainable AI (XAI) component of FIDMF directly enhances the ethical attribute of transparency. By providing human-readable explanations, the system moves beyond 'black-box' decision-making, allowing security analysts to understand and audit the model's behavior. This interpretability is vital for avoiding biased or discriminatory decision-making, which could have serious operational consequences."

Practical deployment guidance

For future real-world deployment, the following guidance outlines key operational procedures. Clients should use the latest global model (M_{global}) asynchronously downloaded from the server for inference. Local training and FedAvg aggregation should be scheduled during low-traffic periods to minimize real-time detection latency. DistilBERT feature enrichment is a one-time preprocessing step, whereas GPT-2 XAI generation must be run on-demand for critical anomalies to avoid constant computational load. The central server must continuously monitor client-side performance metrics (accuracy, loss) to detect model drift or potential poisoning attacks. Finally, periodic local retraining on newly collected data should be performed to ensure adaptability to evolving attack strategies.

Data availability

Data is provided within the manuscript or supplementary information files are stored in git at "<https://github.com/ahmedsalem84/IDS-LLM>".

Received: 17 September 2025; Accepted: 16 February 2026

Published online: 22 February 2026

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Acknowledgements

The authors would like to express their sincere appreciation to the Department of Computer Science, College of Computer and Information Sciences at King Saud University, for providing the academic environment, resources, and continuous support that enabled the successful completion of this research.

Author contributions

Abdullah AlHayan was responsible for the conceptualization of the study, developing the methodology, software implementation, conducting the experiments, and writing the original manuscript. Jalal Al-Muhtadi provided supervision, project administration, validation of the results, and contributed to the critical review and editing of the final manuscript. Both authors have read and approved the final version of the manuscript.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-40763-5>.

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